Coefficient criteria for finite nonnegative linearization

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Abstract. We give two sufficient criteria stated directly in the coefficients of a finite stochastic three-term recurrence. The first assumes that the sum of its diagonal and downward coefficients has positive geometric increments with ratio $q \geq 1$, and that the ratio of those coefficients is constant and nonnegative. A finite algebra construction gives nonnegative linearization, including products beyond the terminating degree. This includes the full positive Krawtchouk range and the dual q-Krawtchouk family with nonconstant increments. The underlying quadratic algebra occurs in Koelink's work on type B Hecke algebras. The second criterion gives explicit perturbation inequalities admitting a relatively open neighborhood of recurrence arrays around every Krawtchouk system in the interior of the positive linearization range. For symmetric finite systems, a quadratic decomposition reduces positivity to two nonsymmetric finite systems. This yields a parity-split criterion covering Szwarc's quadratic system for $p_0 \leq 1/2$ and a q-analogue, and explicit symmetric neighborhoods. For arrays with constant ratio of diagonal to downward coefficients, we prove that nonnegative log-concave Newton coefficients of the profile imply nonnegative linearization, for every size and every product; this inequality criterion contains the Krawtchouk and geometric arrays. None of the criteria requires computing general linearization coefficients.

Keywords. Finite orthogonal polynomials; nonnegative linearization; Krawtchouk polynomials; dual q-Krawtchouk polynomials; quadratic transformation; polynomial hypergroups.

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1. Introduction

Nonnegative linearization (property (P)) asks whether the product of two orthogonal polynomials is a nonnegative linear combination of polynomials in the same basis. For a finite system, the identity is understood on the orthogonality support. This distinction is essential when the formal product degree exceeds the terminal degree. In a normalization with $R_i(1) = 1$, nonnegative coefficients have sum one and define the product probabilities of a finite polynomial hypergroup.

The Krawtchouk case is classical. Eagleson [2] studied positive definite sequences for these polynomials in 1969; Szwarc [6, §5, Problem 2] records the nonnegative-linearization

range as $0 < p_0 \le 1/2$ for binomial weights $\binom{N}{j} p_0^{N-j}(1-p_0)^j$ when normalized at $j = 0$.[1] Our proof below fixes that convention directly from a two-point model. It does not treat Krawtchouk positivity itself as a new result.

Recurrence methods also have a long history. In the second note of his 1969 report, Askey [1, Part II, Theorem 1 and Corollary 1] gives finite recurrence conditions leading to a hyperbolic difference-equation maximum principle. The stated polynomial conclusion concerns products with degree sum at most N. Szwarc's later characterization [7, Theorem 1] treats finite as well as infinite systems through an auxiliary discrete hyperbolic problem. His 2005 survey nevertheless asks for criteria for finite systems, including Krawtchouk and its quadratic system Q_n (Section 7) [6, §5, Problem 2]. Here we seek structural assumptions on the recurrence coefficients that imply positivity for every finite product.

Koelink [3, Theorems 3.1(ii) and 7.8] constructed a triangular quadratic algebra associated with the type B Hecke algebra, and identified the corresponding q-Krawtchouk spherical functions. The dual recurrence in that construction is the geometric family used below. Our algebraic proof extracts its positive multiplication mechanism and expresses the resulting sufficient condition as two scalar patterns in a stochastic recurrence: a constant ratio and geometric increments. No priority is claimed for the algebra or for the resulting named family. For comparison, Miclo [4, Proposition 3.3] gives a spectral characterization of the hypergroup property at a boundary point for irreducible finite birth–death chains by entrywise nonnegativity of products of transition matrices shifted by eigenvalues of their leading principal submatrices.

The second criterion is quantitative stability. Around each Krawtchouk array in the interior of the positive linearization range ($0 < r < 1$, equivalently $0 < p_0 < 1/2$), we exhibit an explicit radius in the space of recurrence coefficients within which all finite products remain nonnegative. Its proof uses one positive term to obtain a reference margin, an error recurrence to control perturbations, and orthogonality to preserve exact zeros outside the triangle support. The resulting neighborhoods contain independently varying coefficient arrays, rather than only a reparametrized classical family. The radius is deliberately conservative and depends on N; no uniform stability radius at the symmetric endpoint is asserted.

Section 2 states the geometric criterion. Sections 3 and 4 give its algebraic proof. Section 5 proves the perturbation inequalities, including explicit nongeometric examples. Section 6 fixes the Krawtchouk normalization and compares the geometric recurrence with Koelink's construction. Section 7 treats Szwarc's quadratic system and inequality hypotheses away from the deepest products. Section 8 proves an inequality criterion for constant-ratio arrays, including all deep products: nonnegative log-concave Newton coefficients of the profile suffice. Section 9 proves that uniformly sampled polynomial deformation rules, and analytic rules whose defining functions extend holomorphically to an open neighborhood of the region Ω specified there, must satisfy two first-order equalities if they keep property (P)

[1] The linked author version, §5, Problem 2, printed p. 125, uses inconsistent parameter conventions: its displayed weights have mean $N(1-p)$, whereas its degree-zero recurrence gives $K_1(x) = (x - pN)/((1-p)N)$. Thus the displayed weights give $\int K_1\, d\mu = (1-2p)/(1-p)$. Corollary 6.1 derives our weights and normalization directly.

at the symmetric Krawtchouk systems, shows that the stability radius of Theorem 5.3 has the optimal order in $1-r$, and records counterexamples to inequality shape conditions at smaller r. The final section lists questions left by these results.

2. Statement and conventions

For integers $j \geq 0$ put $[j]_q = 1 + q + \cdots + q^{j-1}$, with $[0]_q = 0$; in particular $[j]_1 = j$. Consider real coefficients indexed by $0 \leq i \leq N$, where $N \geq 1$, satisfying

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0). \end{gathered} \tag{2.1}$$

Set $R_{-1} = 0$, $R_0 = 1$, and determine $R_1, \ldots, R_N$ by

$$xR_i = a_i R_{i+1} + b_i R_i + c_i R_{i-1} \qquad (0 \leq i < N). \tag{2.2}$$

At $i = 0$, (2.2) gives $R_1 = x$; induction at $x = 1$, using the row sums and $a_i > 0$ for $i < N$, gives $R_i(1) = 1$ for $0 \leq i \leq N$. The terminal relation is $(x - b_N)R_N - c_N R_{N-1} = 0$ on the orthogonality support. Linearization means equality of functions on this support, rather than an unrestricted polynomial identity when the degree sum exceeds N.

Theorem 2.1 *(Geometric increment criterion).* *Assume (2.1). Set $d_i = b_i + c_i$. Suppose*

1. *$b_i/c_i = \beta$ for every $1 \leq i \leq N$, where $\beta \geq 0$;*
2. *$\delta_i = d_i - d_{i-1} > 0$ for $1 \leq i \leq N$;*
3. *for some $q \geq 1$, $\delta_{i+1} = q\delta_i$ whenever $1 \leq i < N$.*

Then the recurrence has a positive orthogonality measure on exactly $N+1$ distinct real points. Its normalized polynomials satisfy

$$R_m R_n = \sum_{k=0}^{N} g(m,n;k) R_k, \qquad g(m,n;k) \geq 0, \qquad \sum_{k=0}^{N} g(m,n;k) = 1 \tag{2.3}$$

on that support for all $0 \leq m, n \leq N$.

Lemma 2.2 *(Coefficient parametrization).* *The hypotheses of Theorem 2.1 are equivalent to*

$$a_i = \frac{q^i [N-i]_q}{[N]_q}, \qquad b_i = (1-r)\frac{[i]_q}{[N]_q}, \qquad c_i = r\frac{[i]_q}{[N]_q} \tag{2.4}$$

for $q \geq 1$, $0 < r \leq 1$. When $N = 1$, one may choose $q = 1$.

Proof. The terminal sum gives $d_N = 1$, and $d_0 = 0$. Geometric increments give $d_i = \delta_1 [i]_q$, hence $\delta_1 = 1/[N]_q$. The constant ratio gives $c_i = d_i/(1+\beta)$ and $b_i = \beta d_i/(1+\beta)$. Put $r = (1+\beta)^{-1}$. Finally $[N]_q - [i]_q = q^i [N-i]_q$. These calculations reverse to give the converse, including $q = 1$ and $N = 1$. ∎

The parameter conventions used below are

q	$q \geq 1$ (geometric increment ratio)
β	$b_i/c_i \geq 0$
r	$1/(1+\beta) \in (0,1]$
p	$1/r = 1+\beta \geq 1$
p_0	$r/(1+r) \in (0,1/2]$
symmetric Krawtchouk case	$r = 1, \quad p_0 = 1/2$

3. A finite algebra with positive multiplication

Fix $p \geq 1$ and $q \geq 1$. Starting with $A_{N+1} = \mathbb{R}$, form recursively the commutative real algebras

$$A_j = A_{j+1}[z_j]/\left(z_j^2 - \left((q-1)\sum_{l=j+1}^{N} z_l + p - 1\right) z_j - pq^{N-j}\right). \tag{3.1}$$

Write $A = A_1$. For $S \subseteq \{1,\dots,N\}$ let $z_S = \prod_{j\in S} z_j$, with $z_\varnothing = 1$.

Lemma 3.1 *(Basis and positivity).* *The elements z_S form a basis of A, and all their multiplication coefficients are nonnegative.*

Proof. Division by a monic quadratic over any commutative coefficient ring gives a unique remainder of degree at most one. Applying this successively in (3.1) gives the asserted basis and dimension 2^N. For positivity, replace a square by

$$z_j^2 = (q-1)\sum_{l>j} z_j z_l + (p-1)z_j + pq^{N-j}.$$

Every displayed coefficient is nonnegative. This rewriting terminates: give z_j weight 2^{N-j} and give a monomial the sum of the weights of its factors, counted with multiplicity. Each replacement strictly decreases this nonnegative integer weight, because $l > j$ implies $2^{N-l} < 2^{N-j}$. The final expression is in the unique squarefree basis, proving the claim. ∎

Lemma 3.2 *(Positive associative inner product).* *Let τ extract the coefficient of $z_\varnothing$. Then*

$$\tau(z_S z_T) = \mathbf{1}_{S=T}\, p^{|S|} q^{\sum_{j\in S}(N-j)}. \tag{3.2}$$

Consequently $\langle u, v\rangle = \tau(uv)$ is positive definite, and multiplication by every element of A is self-adjoint.

Proof. Induct downwards through the tower. If j belongs to exactly one of S, T, their product has the form $z_j v$ with $v \in A_{j+1}$ and has zero constant coefficient in z_j. If it belongs to both, the constant coefficient in z_j is $pq^{N-j} z_{S\setminus\{j\}} z_{T\setminus\{j\}}$ by (3.1). If it belongs to neither, pass to A_{j+1}. Iteration proves (3.2). All diagonal factors are positive. Associativity and commutativity give $\tau((hu)v) = \tau(u(hv))$. ∎

Put $E_k = \sum_{|S|=k} z_S$ for $0 \leq k \leq N$, and set $E_{-1} = E_{N+1} = 0$.

Lemma 3.3 *(Three-term multiplication).* *For $0 \le k \le N$,*

$$E_1 E_k = [k+1]_q E_{k+1} + (p-1)[k]_q E_k + pq^{k-1}[N-k+1]_q E_{k-1}. \tag{3.3}$$

The last term is interpreted as zero for $k = 0$.

Proof. For $k = 0$ the identity reads $E_1 = E_1$. For $N = 1, k = 1$ it is exactly $z_1^2 = (p-1)z_1 + p$. For the induction on N, write F_j for the elementary squarefree sums in $z_2, \dots, z_N$, with $F_j = 0$ outside $0 \le j \le N-1$. Then $E_k = F_k + z_1 F_{k-1}$ and $E_1 = F_1 + z_1$. For $1 \le k \le N-1$, use (3.1) to obtain

$$E_1 E_k = F_1 F_k + z_1 F_k + q z_1 F_1 F_{k-1} + (p-1) z_1 F_{k-1} + pq^{N-1} F_{k-1}.$$

Apply the induction hypothesis to $F_1 F_k$ and $F_1 F_{k-1}$. The coefficients of F_{k+1} and $z_1 F_k$ are respectively $[k+1]_q$ and $1 + q[k]_q = [k+1]_q$. The coefficients of F_k and $z_1 F_{k-1}$ are both $(p-1)[k]_q$, since $1 + q[k-1]_q = [k]_q$. The coefficient of F_{k-1} is

$$pq^{k-1}[N-k]_q + pq^{N-1} = pq^{k-1}[N-k+1]_q,$$

which is also the coefficient of $z_1 F_{k-2}$. Regrouping gives (3.3). For the endpoint $k = N$, $E_N = z_1 F_{N-1}$, so the same square relation gives

$$E_1 E_N = q z_1 F_1 F_{N-1} + (p-1) z_1 F_{N-1} + pq^{N-1} F_{N-1}.$$

The tail endpoint identity is

$$F_1 F_{N-1} = (p-1)[N-1]_q F_{N-1} + pq^{N-2} F_{N-2}.$$

Substitution gives $(p-1)[N]_q E_N + pq^{N-1} E_{N-1}$, as required. This separately checks the terminating boundary. ∎

Proposition 3.4 *(Positive subalgebra).* *The space $B = \operatorname{span}\{E_0, \dots, E_N\}$ is the subalgebra generated by E_1. Every product $E_m E_n$ has nonnegative coefficients in this basis.*

Proof. All E_k are independent, since their squarefree supports are nonempty and disjoint. Equation (3.3) shows that B is invariant under multiplication by E_1. Since $[k+1]_q > 0$, its interior recurrence also expresses each E_{k+1} as a polynomial in E_1. Thus $B = \mathbb{R}[E_1]$, proving closure under multiplication, including products of total index greater than N. Expand such a product as $\sum_k h_k E_k$. Its coefficient at each squarefree monomial z_S is $h_{|S|}$. Lemma 3.1 makes that coefficient nonnegative. There is a subset of each cardinality $0, \dots, N$, so every h_k is nonnegative. ∎

4. Normalization and the finite orthogonal system

Lemma 4.1 *(Positive normalization).* *Evaluation $z_j \mapsto pq^{N-j}$ defines an algebra character ϵ of A. Put $t_k = \epsilon(E_k) > 0$. Then*

$$t_0 = 1, \quad t_1 = p[N]_q, \qquad \frac{t_{k+1}}{t_k} = pq^k \frac{[N-k]_q}{[k+1]_q} \quad (0 \le k < N). \tag{4.1}$$

Proof. For $h = N - j$, substitution in the right-hand side of (3.1) gives

$$(q-1)p^2q^h[h]_q + (p-1)pq^h + pq^h \;=\; p^2q^h\big((q-1)[h]_q + 1\big) \;=\; p^2q^{2h}.$$

Thus all defining relations hold. The generating polynomial of the t_k is $T(u) = \prod_{j=0}^{N-1}(1 + pq^j u)$, whose coefficients are strictly positive. When $q > 1$, the identity $(1+pu)T(qu) = (1+pq^N u)T(u)$ gives $(q^k-1)t_k = p(q^N - q^{k-1})t_{k-1}$ for $1 \le k \le N$, which is (4.1). When $q = 1$, $t_k = \binom{N}{k}p^k$ and the same ratio follows directly. ∎

Proof of Theorem 2.1. Use Lemma 2.2 and put $p = 1/r \ge 1$. Construct A, B as above and set $X = E_1/t_1$, $U_k = E_k/t_k$. Divide (3.3) by t_1t_k and use (4.1). The three coefficients are

$$\begin{aligned}
\frac{[k+1]_q t_{k+1}}{p[N]_q t_k} &= \frac{q^k[N-k]_q}{[N]_q},\\
\frac{(p-1)[k]_q}{p[N]_q} &= (1-r)\frac{[k]_q}{[N]_q},\\
\frac{pq^{k-1}[N-k+1]_q t_{k-1}}{p[N]_q t_k} &= r\frac{[k]_q}{[N]_q}.
\end{aligned}$$

At $k = 0$ omit the third expression; at $k = N$ omit the first. Consequently $U_k = R_k(X)$ for every $k \le N$, with the stated terminal relation.

The orthogonality support follows from the real spectral theorem in finite dimensions. On B use the positive inner product from Lemma 3.2. Multiplication by X is self-adjoint. The vector $1 = E_0$ is cyclic because the E_k are polynomials of successive degrees in X and form a basis. In an orthogonal eigenspace decomposition of a self-adjoint operator, the cyclic span of one vector has dimension at most the number of distinct eigenvalues. Here the cyclic span has dimension $N+1$, so there are exactly $N+1$ distinct eigenvalues, and the projection of 1 on every eigenspace is nonzero. Give each eigenvalue the squared norm of that projection as its mass. All masses are positive, sum to $\langle 1, 1\rangle = 1$, and yield

$$\int f(x)h(x)\,d\mu(x) \;=\; \langle f(X), h(X)\rangle$$

for real polynomials f, h, identifying an algebra element with its action on 1. The E_k are orthogonal by (3.2), so the R_k are orthogonal for this measure.

Proposition 3.4 and positive t_k give nonnegative product coefficients for the U_k. Applying the character ϵ gives their sum equal to one because $\epsilon(U_k) = 1$. Polynomial identities in the algebra hold on every spectral point; conversely the R_k, having successive degrees at most N, form a basis of functions on the $N+1$ distinct points. This proves uniqueness and (2.3) for all products. ∎

5. A criterion for small coefficient perturbations

Write the tridiagonal recurrence matrix with the row convention

$$J_{i,i+1} \;=\; a_i, \qquad J_{i,i} \;=\; b_i, \qquad J_{i,i-1} \;=\; c_i,$$

with all other entries zero and indices outside $0, \dots, N$ omitted.

Lemma 5.1 *(Finite recurrence realization and triangle support).* *Every recurrence satisfying (2.1) has a positive orthogonality measure on exactly $N+1$ real points. If J is its tridiagonal recurrence matrix, then its finite product coefficients satisfy*

$$g(m,n;k) \;=\; [R_m(J)]_{n,k}. \tag{5.1}$$

They vanish unless $|m-n| \le k \le \min(m+n, N)$, regardless of their signs inside this interval.

Proof. Choose positive π_i with $\pi_0 = 1$ and $\pi_{i+1} = \pi_i a_i / c_{i+1}$. For $D = \operatorname{diag}(\sqrt{\pi_i})$, $H = DJD^{-1}$ is real symmetric. Induction in (2.2) gives the row-vector identity $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$, hence $e_0^{\mathsf{T}} R_i(H) = \pi_i^{-1/2} e_i^{\mathsf{T}}$. In particular e_0 is cyclic for H. The same finite spectral argument as in Theorem 2.1 gives $N+1$ distinct eigenvalues and positive spectral masses at e_0, and

$$\int R_i R_j \, d\mu \;=\; e_0^{\mathsf{T}} R_i(H) R_j(H) e_0 \;=\; \mathbf{1}_{i=j} \pi_i^{-1}.$$

The terminal relation holds on this support: its polynomial $Q(x) = (x - b_N) R_N - c_N R_{N-1}$ satisfies $e_0^{\mathsf{T}} Q(J) = 0$. Since e_0 is cyclic and $Q(J)$ commutes with J, $Q(J) = 0$. Multiplying a basis polynomial by x acts by the corresponding row of J, including the terminal row. Induction on powers of x therefore gives (5.1). Orthogonality also gives

$$g(m,n;k) \;=\; \frac{\int R_m R_n R_k \, d\mu}{\int R_k^2 \, d\mu}.$$

If one of m, n, k exceeds the sum of the other two, its polynomial is orthogonal to their product, whose degree is smaller. This proves the asserted support. ∎

Lemma 5.2 *(A uniform positive margin in the asymmetric case).* *For the Krawtchouk recurrence (2.4) with $q = 1$ and $0 < r < 1$, put $s = \min(r, 1-r)$. Every coefficient in its triangle support obeys*

$$g(m,n;k) \;\ge\; (s/4)^N \quad \text{if } |m-n| \le k \le \min(m+n, N). \tag{5.2}$$

Proof. Specialize the algebra above to $q = 1$, and put $y_i = z_i/p$, where $p = 1/r$. Then $y_i^2 = (1-r) y_i + r$, and $U_m = \binom{N}{m}^{-1} \sum_{|S|=m} y_S$. The reduction of a product $y_S y_T$ has nonnegative coefficients, each chosen term being a product of at most N factors from $\{r, 1-r\}$. Its coefficient is therefore at least s^N. For each admissible k, at least one such term has degree k. Assume $m \ge n$. If $k \le m$, take $T \subseteq S$, with sizes n, m, and retain the linear term for exactly $k - (m-n)$ overlap coordinates. This number lies between 0 and n. If $k \ge m$, choose sets with overlap $m+n-k$ and union of size $k \le N$, and retain the linear term at every overlap coordinate. Permutation symmetry makes the coefficient of y_U depend only on $|U| = k$. Hence the coefficient in the normalized elementary-symmetric basis is at least

$$\frac{s^N \binom{N}{k}}{\binom{N}{m}\binom{N}{n}} \;\ge\; \frac{s^N}{4^N}.$$

This uses a single positive term and a denominator bound, not an explicit formula for the product coefficients. ∎

Theorem 5.3 *(Explicit perturbation inequalities).* *Let $N \geq 1$, $0 < r < 1$, $s = \min(r, 1-r)$, and*

$$\eta_N(r) = \frac{1}{2}\left(\frac{s}{32N}\right)^N. \tag{5·3}$$

Suppose coefficients $\widetilde{a}_i, \widetilde{b}_i, \widetilde{c}_i$ satisfy (2.1), with $\widetilde{b}_i \geq 0$ for every i, and

$$\max_{0\leq i\leq N}\left\{\left|\widetilde{a}_i - \frac{N-i}{N}\right|, \left|\widetilde{b}_i - \frac{(1-r)i}{N}\right|, \left|\widetilde{c}_i - \frac{ri}{N}\right|\right\} \leq \eta_N(r). \tag{5·4}$$

Then the finite system has nonnegative linearization. Throughout its triangle support,

$$\widetilde{g}(m,n;k) \geq \tfrac{1}{2}(s/4)^N.$$

The product coefficients sum to one.

Proof. Use tildes for the perturbed system and untilded letters for its $q = 1$ reference at r. Let $\eta = \eta_N(r)$ and use the maximum absolute row-sum matrix norm. Both J and $\widetilde{J}$ are stochastic matrices, so $\|J\|_\infty = \|\widetilde{J}\|_\infty = 1$ and $\|\widetilde{J} - J\|_\infty \leq 3\eta$. Put $B_m = R_m(J)$, $\widetilde{B}_m = \widetilde{R}_m(\widetilde{J})$ and $e_m = \|\widetilde{B}_m - B_m\|_\infty$. Theorem 2.1 at $q = 1$ and (5.1) give $\|B_m\|_\infty = 1$ for all $m \leq N$. Also $e_0 = 0$ and $e_1 \leq 3\eta$.

Subtract the two matrix recurrences for $1 \leq m < N$. Writing $\Delta_m = \widetilde{B}_m - B_m$ gives exactly

$$\begin{aligned}\widetilde{a}_m\Delta_{m+1} = {} & (\widetilde{J} - \widetilde{b}_m I)\Delta_m - \widetilde{c}_m\Delta_{m-1} \\ & + \big(\widetilde{J} - J - (\widetilde{b}_m - b_m)I\big)B_m \\ & - (\widetilde{c}_m - c_m)B_{m-1} - (\widetilde{a}_m - a_m)B_{m+1}.\end{aligned}$$

Since $\eta \leq 1/(2N)$, $\widetilde{a}_m \geq 1/(2N)$. The terms not containing a Δ have combined norm at most 6η. Also $0 \leq \widetilde{b}_m, \widetilde{c}_m \leq 1$, so

$$e_{m+1} \leq 4Ne_m + 2Ne_{m-1} + 12N\eta. \tag{5·5}$$

Set $L = 8N$. Induction gives $e_m \leq \eta L^m$ for $1 \leq m \leq N$: the base case follows from $3 \leq L$, and, for $m \geq 1$,

$$4N + \frac{2N}{L} + \frac{12N}{L^m} \leq 4N + \frac{1}{4} + \frac{3}{2} \leq 8N = L.$$

Thus every matrix entry differs from its reference by at most

$$\eta(8N)^N = \frac{1}{2}(s/4)^N.$$

Lemma 5.2 gives the positive lower bound on triangle entries, while Lemma 5.1 makes all entries outside the triangle exactly zero. Finally $\widetilde{R}_m(1) = 1$ follows by recurrence from the stochastic coefficient sums, so evaluating each product at 1 gives coefficient sum one. Here 1 belongs to the support because $\widetilde{J}\mathbf{1} = \mathbf{1}$ and Lemma 5.1 identifies that support with its spectrum. No product-positivity inequality for the perturbed system has been assumed. ∎

Corollary 5.4 *(Finite systems outside the geometric family).* *For every $N \geq 2$, take the Krawtchouk coefficients at $r = 1/2$, and replace only*

$$a_1 \longmapsto a_1 - h, \qquad b_1 \longmapsto b_1 + h, \qquad 0 < h \leq \frac{1}{4}(64N)^{-N}.$$

These recurrences have nonnegative linearization and do not satisfy the constant-ratio hypothesis of Theorem 2.1. More generally, Theorem 5.3 admits a relatively open neighborhood of the asymmetric Krawtchouk array in the affine space defined by the row-sum and boundary equalities in (2.1).

Proof. All initial and terminal conventions and stochastic sums are preserved. The bound on h is half the radius (5.3) at $r = 1/2$, and is smaller than $a_1 = (N-1)/N$, so all required signs hold. The perturbation theorem applies. At index 1, $b_1/c_1 = 1 + 2Nh > 1$, whereas at index N the ratio remains 1; these indices differ because $N \geq 2$. Thus this array lies outside the geometric family. At the asymmetric reference, all free coefficients constrained to be positive are strictly positive; the strict version of (5.4) therefore contains a relatively open neighborhood in that affine space; the sign conditions define the admissible region within it. ∎

6. Applications and scope

Corollary 6.1 *(Full positive Krawtchouk range).* *For every $N \geq 1$ and $0 < p_0 \leq 1/2$, Krawtchouk polynomials orthogonal for binomial weights $\binom{N}{j} p_0^{N-j}(1-p_0)^j$ have nonnegative linearization when normalized at $j = 0$.*

Proof. To fix the convention without relying on a named recurrence, put $r = p_0/(1-p_0) \in (0, 1]$, and let $Y_1, \ldots, Y_N$ be independent variables with values $1, -r$ of probabilities $p_0, 1 - p_0$. Their mean is zero and $Y_i^2 = (1-r)Y_i + r$. The normalized elementary symmetric sums

$$K_i(j) = \binom{N}{i}^{-1} \sum_{|S|=i} \prod_{l \in S} Y_l$$

depend only on the number j of negative coordinates. They are orthogonal, because the expectation of two distinct squarefree monomials is zero: some independent mean-zero variable occurs once. Their weights in j are exactly the stated binomial weights. With $x = N^{-1} \sum_l Y_l = 1 - j/(N(1-p_0))$, multiplication by x gives

$$xK_i = \frac{N-i}{N} K_{i+1} + (1-r)\frac{i}{N} K_i + r\frac{i}{N} K_{i-1}.$$

Indeed, in multiplying the elementary sum by $\sum_l Y_l$, indices outside a subset give $(i+1)$ copies of each subset of size $i+1$; the linear part of Y_l^2 gives $(1-r)i$ copies of each subset of size i; and the constant part gives $r(N-i+1)$ copies of each subset of size $i-1$. Division by the binomial normalizers gives the displayed recurrence, including $i = N$. Its leading coefficient in x is positive, its degree is i, and its value at $j = 0$ is one. These properties identify the normalized orthogonal polynomials for the binomial measure. Apply

Theorem 2.1 with $q = 1$. The range is sharp in this endpoint normalization: if $1/2 < p_0 < 1$, then $r = p_0/(1-p_0) > 1$ and $g(1,1;1) = b_1 = (1-r)/N < 0$. For $N = 1$, this coefficient follows from the terminal relation. ■

Corollary 6.2 *(The dual q-Krawtchouk case).* *For every $N \geq 2$, $q > 1$, and $0 < r \leq 1$, the recurrence (2.4) has nonnegative linearization. In the normalization $R_0 = 1$, $R_1 = x$, it is outside the Krawtchouk recurrence class.*

Proof. Its increments are $d_i - d_{i-1} = q^{i-1}/[N]_q$, so all hypotheses hold. For a Krawtchouk recurrence in this normalization the increments of $b_i + c_i$ are constant. Here their ratio is $q > 1$ and $N \geq 2$, so they are not constant. An affine change of the support transporting the same endpoint-normalized basis leaves the multiplication coefficients in the basis unchanged; it cannot remove this distinction. ■

Proposition 6.3 *(Relation to the existing algebraic construction).* *The quadratic relations (3.1) are those of Koelink [3, Theorem 3.1(ii), equation (3.3)], with $z_j = u(x^j)$ and the same parameters p, q. The recurrence in this paper is the normalized recurrence of his Theorem 7.1, with $r = 1/p$; its polynomial-in-the-spectral-variable interpretation is the dual q-Krawtchouk interpretation described after his Theorem 7.8.*

Proof. The source square relation is $u(x^j)^2 = (q-1)u(x^j)\sum_{l>j} u(x^l) + (p-1)u(x^j) + pq^{N-j}$, which is (3.1). In Theorem 7.1 the coefficients of the next, current and previous indices are respectively $pq^i(1-q^{N-i})$, $(p-1)(1-q^i)$ and $1-q^i$. Division by $p(1-q^N)$ gives (2.4) for $q > 1$. For $0 \leq f \leq N$, the corresponding spectral coordinate is

$$x_f = \frac{p(1-q^{N-f}) - (1-q^f)}{p(1-q^N)} = \frac{q^{N-f} - rq^f + r - 1}{q^N - 1}, \qquad r = 1/p.$$

Writing $\phi_f(w_d)$ for Koelink's spherical-function values, we have $R_d(x_f) = \phi_f(w_d)$ for $0 \leq d \leq N$: the initial value is one and the normalized recurrences agree, so induction in d identifies them. As $q \to 1$, dividing numerator and denominator by $q-1$ and using the finite geometric sums gives $x_f \to 1 - (1+r)f/N$. The $q = 1$ case follows directly from the construction above, without division by zero. His discussion after Theorem 7.8 explicitly interchanges polynomial degree and spectral index to obtain the dual family. ■

The geometric criterion covers the full positive Krawtchouk range, including the symmetric endpoint. The perturbation criterion allows freely varying coefficients around each Krawtchouk array in the interior of the positive linearization range; its stated radius is conservative and decreases with N. Both hypotheses can be checked using a number of arithmetic comparisons proportional to N, without forming any general product coefficients or testing polynomial matrices. The proofs use positive algebra multiplication, a lower bound from a single positive term, and a norm estimate. They provide sufficient conditions, not a characterization of all finite systems with property (P). The geometric parameter family and underlying algebra predate Szwarc's survey; this paper makes no claim of historical priority for these constructions or criteria.

7. Szwarc's quadratic system

Problem 2 of [6, §5] does not stop at the Krawtchouk recurrence. With his parameter written as p_0, Szwarc introduces the symmetric recurrence of length $2N+1$

$$yQ_{2n} = (N-n)Q_{2n+1} + nQ_{2n-1}, \qquad yQ_{2n-1} = (1-p_0)Q_{2n} + p_0 Q_{2n-2}, \tag{7.1}$$

records that $Q_{2n}(y)$ is the Krawtchouk polynomial $K_n(y^2)$ of his recurrence, states that the Q_n have nonnegative linearization for $p_0 \le 1/2$, and asks for a general criterion that would handle this case. His Theorem 3.3 for infinite symmetric systems does not apply. In the normalization (7.1) its monotonicity hypotheses hold, and its hypothesis $\alpha_n \le \gamma_n$ holds at odd indices exactly when $p_0 \le 1/2$; at an even index $2n$ it reads $n \le N-n$ and fails in the upper half. Some failure of this kind is forced in every finite system, whose terminal upward coefficient vanishes. This section proves the positivity of (7.1) by a graded version of the algebra of Section 3, and then gives an explicit coefficient criterion for symmetric finite systems near (7.1).

Put $M = 2N$, $R_{2n}(t) = Q_{2n}(\sqrt{N}t)$ and $R_{2n-1}(t) = \sqrt{N}\,Q_{2n-1}(\sqrt{N}t)$. Substitution in (7.1) shows that the R_k satisfy (2.2), with N replaced by M, for the array

$$b_k = 0, \qquad (a_{2n}, c_{2n}) = \Big(\frac{N-n}{N}, \frac{n}{N}\Big), \qquad (a_{2n-1}, c_{2n-1}) = (1-p_0,\, p_0). \tag{7.2}$$

It satisfies (2.1) with M in place of N whenever $0 < p_0 < 1$. The two normalizations differ by positive factors and a positive dilation, so they have the same product coefficients up to positive factors.

Theorem 7.1 *(Szwarc's quadratic system).* *Let $N \ge 1$ and $0 < p_0 \le 1/2$. The array (7.2) has a symmetric positive orthogonality measure on $2N+1$ points, and for all $0 \le m, n \le 2N$*

$$R_m R_n = \sum_{k=0}^{2N} g(m,n;k) R_k, \qquad g(m,n;k) \ge 0, \qquad \sum_k g(m,n;k) = 1$$

on that support. If $1/2 < p_0 < 1$, then $g(2,2;2) < 0$.

Throughout this section put $r = p_0/(1-p_0)$, so that $0 < r \le 1$ exactly when $p_0 \le 1/2$. Let A be the commutative algebra generated by $y_1, \dots, y_N$ subject to $y_j^2 = (1-r)y_j + r$. This is the algebra of Section 3 with $q = 1$ after the rescaling $y_j = z_j/p$ used in Lemma 5.2. Its squarefree monomials y_S form a basis, and for $r \le 1$ all multiplication coefficients are nonnegative. Call an element of A nonnegative if its coefficients in this basis are nonnegative.

Lemma 7.2 *(Edge idempotents).* *For $1 \le l \le N$ put $f_l = (r + y_l)/(1+r)$. Then $y_l f_l = f_l$ and $f_l^2 = f_l$. The ideal $f_l A$ has basis $f_l y_S$, $S \subseteq \{1, \dots, N\} \setminus \{l\}$, and $f_l u$ has nonnegative coefficients in this basis whenever $u \in A$ is nonnegative.*

Proof. First $(1+r)y_l f_l = r y_l + (1-r)y_l + r = (1+r)f_l$. Then $(1+r)f_l^2 = r f_l + y_l f_l = (1+r)f_l$. Let A' be generated by the y_j with $j \ne l$, so that $A = A' \oplus y_l A'$. Since $f_l y_l = f_l$, $f_l A = f_l A'$. The map $u \mapsto f_l u = (ru + y_l u)/(1+r)$ is injective on A' by the direct sum, which gives the basis. For the last claim, $f_l y_S = f_l y_{S \setminus \{l\}}$ for every S. ∎

Let $V = A \oplus \bigoplus_{l=1}^{N} f_l A$, on which A acts by multiplication in every summand, and call a vector nonnegative if all of its coordinates are nonnegative in the bases above. Define the A-linear operator

$$Y(u; v_1, \dots, v_N) = \Big(\frac{1}{N} \sum_{l=1}^{N} v_l \,;\, f_1 u, \dots, f_N u \Big).$$

Put $U_{2n} = \binom{N}{n}^{-1} \sum_{|S|=n} y_S \in A$ for $0 \le n \le N$. For $0 \le n \le N-1$, let U_{2n+1} be the vector in $\bigoplus_l f_l A$ whose lth coordinate is $\binom{N-1}{n}^{-1} \sum f_l y_S$, summed over $S \subseteq \{1, \dots, N\} \setminus \{l\}$ with $|S| = n$.

Lemma 7.3 *(Three-term action).* *With $U_{-1} = U_{2N+1} = 0$ and the coefficients (7.2), $Y U_k = a_k U_{k+1} + c_k U_{k-1}$ for $0 \le k \le 2N$.*

Proof. The lth coordinate of $Y \sum_{|S|=n} y_S$ is $\sum_{S \not\ni l} f_l y_S + \sum_{S' \not\ni l} f_l y_{S'}$, with $|S| = n$ and $|S'| = n-1$, because $f_l y_S = f_l y_{S \setminus \{l\}}$. Dividing by $\binom{N}{n}$ and using $\binom{N-1}{n}/\binom{N}{n} = (N-n)/N$ and $\binom{N-1}{n-1}/\binom{N}{n} = n/N$ gives the even case. For the odd case, $(1+r) f_l y_S = r y_S + y_{S \cup \{l\}}$ when $l \notin S$. Each n-set occurs for $N-n$ choices of l and each $(n+1)$-set for $n+1$ choices, so

$$Y U_{2n+1} = \frac{r(N-n)\binom{N}{n} U_{2n} + (n+1)\binom{N}{n+1} U_{2n+2}}{N(1+r)\binom{N-1}{n}} = \frac{r U_{2n} + U_{2n+2}}{1+r}.$$

Finally $r/(1+r) = p_0$ and $1/(1+r) = 1 - p_0$. ∎

Proof of Theorem 7.1. Lemma 5.1 gives a positive measure on the $2N+1$ zeros of the terminal polynomial $T(t) = t R_{2N}(t) - R_{2N-1}(t)$. Since $b \equiv 0$, the matrix J is conjugate to $-J$ by $\operatorname{diag}((-1)^i)$, which fixes e_0; hence the measure is symmetric and $R_k(-t) = (-1)^k R_k(t)$. Put $\Phi(f) = f(Y) U_0$, where $U_0 = (1; 0, \dots, 0)$. Lemma 7.3 gives $\Phi(R_k) = U_k$ for $k \le 2N$ and $\Phi(T) = Y U_{2N} - U_{2N-1} = 0$. Every polynomial vanishing on the support is a multiple of T, so Φ maps it to zero. Hence the product identity on the support becomes

$$R_m(Y) R_n(Y) U_0 = \sum_k g(m, n; k) U_k.$$

The supports of the U_k in the bases of V are nonempty and disjoint: U_{2n} uses the y_S with $|S| = n$, and U_{2n+1} uses the $f_l y_S$ with $l \notin S$, $|S| = n$. Each basis vector lies in exactly one of them, with positive coefficient. Consequently $g(m, n; k) \ge 0$ for every k whenever the left side is nonnegative.

Suppose m is even. Since polynomials in Y commute and Y is A-linear,

$$R_m(Y) R_n(Y) U_0 = R_n(Y)(U_m U_0) = U_m \, R_n(Y) U_0 = U_m U_n.$$

Here $U_m \in A$ and U_n are nonnegative, so their product is nonnegative by the positivity of A and Lemma 7.2. The same holds when n is even, since $g(m, n; k) = g(n, m; k)$. If m, n are odd and k is odd, then $g(m, n; k) = 0$ by symmetry of the measure. If m, n are odd and k is even, then $g(m, n; k) \int R_k^2 \, d\mu = \int R_m R_n R_k \, d\mu = g(k, m; n) \int R_n^2 \, d\mu \ge 0$. The point

$t = 1$ belongs to the support because $J\mathbf{1} = \mathbf{1}$, and $R_k(1) = 1$, so the coefficients sum to one.

For $1/2 < p_0 < 1$ we have $r > 1$. The identities above are purely algebraic and remain valid, but A is no longer positive. In particular $U_2^2 = N^{-2}\big((1-r)\sum_j y_j + Nr + 2\sum_{i<j} y_i y_j\big)$, whose U_2 coefficient $g(2,2;2) = (1-r)/N$ is negative. ∎

Since Y^2 acts on A by multiplication by $N^{-1}\sum_l f_l = (r + X)/(1 + r)$, with $X = N^{-1}\sum_l y_l$, the even part of the theorem is Corollary 6.1: $R_{2n}(t) = K_n\big((1+r)t^2 - r\big)$ in the normalization of that corollary. Thus Theorem 7.1 recovers the full positive Krawtchouk range by Szwarc's quadratic route. At $p_0 = 1/2$ the operator Y is the normalized adjacency operator of the subdivided N-cube seen from a vertex, and the theorem reduces to counting in that graph.

Lemma 7.4 *(Margin for the quadratic system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1-p_0)$ and $s = \min(r, 1-r)$. For the array (7.2),*

$$g(m,n;k) \geq (rs/8)^N \qquad \text{whenever } |m-n| \leq k \leq m+n \text{ and } m+n+k \text{ is even.}$$

All other coefficients vanish.

Proof. The vanishing statements are Lemma 5.1 and the symmetry used above. We use $\binom{N}{k} \leq 2^N$ throughout. Up to the order of m, n, there are three cases.

If m, n, k are even, the coefficient is a Krawtchouk coefficient computed in A, so Lemma 5.2 gives the lower bound $(s/4)^N \geq (rs/8)^N$.

Let $m = 2a$, $n = 2b+1$ and $k = 2c+1$, so that

$$\max(a-b-1,\, b-a) \leq c \leq \min(a+b,\, N-1).$$

Fix l and a set $U \not\ni l$ with $|U| = c$. The lth coordinate of $U_m U_n$ is

$$\binom{N}{a}^{-1}\binom{N-1}{b}^{-1} \sum_{|T|=a} \sum_{S \not\ni l,\ |S|=b} f_l\, y_{T\setminus\{l\}}\, y_S.$$

Every nonzero term arising from a chosen reduction of $y_{T\setminus\{l\}} y_S$ is a product of at most $N-1$ factors from $\{r, 1-r\}$, and therefore has coefficient at least s^{N-1}. Put $a' = |T \setminus \{l\}|$, which can equal $a-1$ when $a \geq 1$ and a when $a \leq N-1$. In the $N-1$ coordinates other than l, sets of sizes a' and b with overlap o, retaining the linear term on $t \leq o$ overlap coordinates, give monomials of size $a' + b - 2o + t$. As o ranges over its admissible values these sizes fill the interval $\big[|a'-b|, \min(a'+b, N-1)\big]$. The two admissible values of a' cover every c above. Hence $f_l y_U$ has coefficient at least $s^{N-1}\binom{N}{a}^{-1}\binom{N-1}{b}^{-1}$, and comparison with U_k gives

$$g(m,n;k) \geq \binom{N-1}{c} \frac{s^{N-1}}{\binom{N}{a}\binom{N-1}{b}} \geq \frac{2s^{N-1}}{4^N} \geq \Big(\frac{s}{4}\Big)^N.$$

Finally let $m = 2b+1$, $n = 2c+1$ and $k = 2a$. With π_i as in the proof of Lemma 5.1, (7.2) gives $\pi_{2a} = \binom{N}{a} r^{-a}$ and $\pi_{2c+1} = \binom{N-1}{c} r^{-c}/p_0$. Orthogonality gives

$$g(2b+1, 2c+1; 2a) = g(2a, 2b+1; 2c+1)\frac{\pi_{2a}}{\pi_{2c+1}} \geq \frac{2s^{N-1}}{4^N} \cdot \frac{p_0 r^N}{2^{N-1}} \geq \frac{2s^{N-1} r^{N+1}}{8^N},$$

using $r^{c-a} \geq r^N$ and $p_0 \geq r/2$. Since $s \leq r$, the last quantity is at least $(rs/8)^N$. ∎

The proof of Theorem 5.3 uses only three properties of its reference: nonnegative linearization with sums one, a lower bound for the upward coefficients, and a margin on the known support. We record it in that generality.

Lemma 7.5 *(Stability of a reference with a margin).* *Let a, b, c satisfy (2.1) on the indices $0, \dots, M$, have nonnegative linearization, and satisfy $a_i \geq \alpha$ for $i < M$, where $0 < \alpha \leq 1$. Suppose $g(m, n; k) \geq \delta > 0$ on a set Σ of index triples. Let $\widetilde{a}, \widetilde{b}, \widetilde{c}$ satisfy (2.1), with $\widetilde{b}_i \geq 0$, and differ from a, b, c by at most η in every entry. If $\eta \leq \alpha/2$ and $\eta(8/\alpha)^M \leq \delta/2$, then $|\widetilde{g}(m, n; k) - g(m, n; k)| \leq \delta/2$ for all m, n, k; in particular $\widetilde{g} \geq \delta/2$ on Σ.*

Proof. Repeat the proof of Theorem 5.3 with N replaced by M. Now $\widetilde{a}_m \geq \alpha/2$, so the subtracted recurrences give $e_{m+1} \leq (4e_m + 2e_{m-1} + 12\eta)/\alpha$. With $L = 8/\alpha \geq 8$, induction gives $e_m \leq \eta L^m$: the base case is $e_1 \leq 3\eta$, and the step needs $4L^m + 2L^{m-1} + 12 \leq 8L^m$, which holds for $m \geq 1$. Every entry of $\widetilde{B}_m - B_m$ is therefore at most $\eta L^M \leq \delta/2$, and (5.1) identifies these entries with the product coefficients. ∎

Theorem 5.3 is the case $M = N$, $\alpha = 1/N$ and $\delta = (s/4)^N$ of this lemma.

Theorem 7.6 *(Symmetric criterion near Szwarc's system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1 - p_0)$, $s = \min(r, 1 - r)$, and*

$$\eta = \frac{1}{2}\Big(\frac{rs}{8}\Big)^N (16N)^{-2N}.$$

Let c_k be the downward coefficients in (7.2). Take real numbers $\widetilde{c}_1, \dots, \widetilde{c}_{2N-1}$ with $|\widetilde{c}_k - c_k| \leq \eta$, and put $\widetilde{c}_0 = 0$, $\widetilde{c}_{2N} = 1$, $\widetilde{b}_k = 0$ and $\widetilde{a}_k = 1 - \widetilde{c}_k$. The resulting symmetric finite system has nonnegative linearization with sums one. Moreover $\widetilde{g}(m, n; k) \geq \frac{1}{2}(rs/8)^N$ whenever $|m - n| \leq k \leq m + n$ and $m + n + k$ is even, and $\widetilde{g}(m, n; k) = 0$ otherwise.

Proof. The perturbed array satisfies (2.1) with $M = 2N$: $\widetilde{a}_k \geq 1/N - \eta > 0$ for even $k < 2N$, $\widetilde{a}_k \geq 1/2 - \eta > 0$ for odd k, and $\widetilde{c}_k \geq \min(p_0, 1/N) - \eta > 0$ for $k > 0$. Theorem 7.1 shows that the reference has nonnegative linearization, and $a_k \geq 1/(2N)$ for $k < 2N$. Lemma 7.4 gives the margin $\delta = (rs/8)^N$ on the stated set. Apply Lemma 7.5 with $\alpha = 1/(2N)$: then $\eta \leq \alpha/2$ and $\eta(16N)^{2N} = \delta/2$. Off the stated set, $\widetilde{g}$ vanishes by Lemma 5.1 and by the symmetry of the perturbed measure, which holds because $\widetilde{b} \equiv 0$. The sums equal one because $\widetilde{R}_m(1) = 1$ and 1 lies in the support. ∎

Corollary 7.7 *(Symmetric systems outside both families).* *For $N \geq 2$, $0 < p_0 < 1/2$ and $0 < h \leq \eta$, the array obtained from (7.2) by $c_1 \mapsto p_0 + h$, $a_1 \mapsto 1 - p_0 - h$ has nonnegative linearization. It is neither of the form (7.2) nor of the form (2.4). More generally, Theorem 7.6 admits a relatively open set of symmetric stochastic arrays, with $2N - 1$ free coefficients, around each array (7.2) with $0 < p_0 < 1/2$.*

Proof. Theorem 7.6 applies. The odd downward coefficients $c_1 = p_0 + h$ and $c_3 = p_0$ differ, so the array is not of the form (7.2); here $N \geq 2$ ensures $3 < 2N$. In (2.4), on $2N + 1$

indices, the downward coefficients $c_i = r[i]_q/[2N]_q$ are strictly increasing, whereas $c_3 < c_1$. ∎

A criterion for all symmetric finite systems. Szwarc uses the quadratic transformation [6, §3.5] to pass from a symmetric system to its even part. For finite systems the converse direction also works: positivity of a symmetric system follows from positivity of two nonsymmetric finite systems, both read off from its coefficients. Any criterion for nonsymmetric finite systems therefore becomes a criterion for symmetric ones.

Theorem 7.8 *(Quadratic decomposition).* *Let $N \geq 2$, and let a_k, c_k $(0 \leq k \leq 2N)$ with $b \equiv 0$ satisfy (2.1) with $2N$ in place of N. Put $a_k = c_k = 0$ for $k \notin [0, 2N]$, and define the even component*

$$\hat{a}_n = a_{2n}a_{2n+1}, \quad \hat{b}_n = c_{2n}a_{2n-1} + a_{2n}c_{2n+1}, \quad \hat{c}_n = c_{2n}c_{2n-1} \qquad (0 \leq n \leq N)$$

and the odd component

$$\breve{a}_n = a_{2n+1}a_{2n+2}, \quad \breve{b}_n = a_{2n+1}c_{2n+2} + c_{2n+1}a_{2n}, \quad \breve{c}_n = c_{2n+1}c_{2n} \qquad (0 \leq n < N).$$

After the affine normalization $(\hat{a}_n, \hat{b}_n, \hat{c}_n) \mapsto (\hat{a}_n, \hat{b}_n - \hat{b}_0, \hat{c}_n)/\hat{a}_0$, and likewise for the odd component, these arrays satisfy (2.1) on the indices $0, \dots, N$ and $0, \dots, N-1$. If both have nonnegative linearization, so does the symmetric array. More precisely, with $\hat{g}$ and $\breve{g}$ the product coefficients of the components and terms with indices out of range omitted,

$$g(2a, 2b; 2c) = \hat{g}(a, b; c), \qquad g(2a, 2b+1; 2c+1) = a_{2a}\,\breve{g}(a, b; c) + c_{2a}\,\breve{g}(a-1, b; c),$$

the coefficients with $m+n+k$ odd vanish, and $g(2a+1, 2b+1; 2c) \int R_{2c}^2\, d\mu = g(2c, 2a+1; 2b+1) \int R_{2b+1}^2\, d\mu$. Conversely, nonnegative linearization of the symmetric array implies that of its even component.

Proof. By symmetry $R_{2n}(t) = P_n(t^2)$ and $R_{2n+1}(t) = tL_n(t^2)$ for polynomials P_n, L_n of degree n. Applying (2.2) twice gives, with $x = t^2$,

$$xP_n = \hat{a}_n P_{n+1} + \hat{b}_n P_n + \hat{c}_n P_{n-1}, \qquad xL_n = \breve{a}_n L_{n+1} + \breve{b}_n L_n + \breve{c}_n L_{n-1}.$$

The row sums of the components equal $a_{2n} + c_{2n} = 1$ and $a_{2n+1} + c_{2n+1} = 1$; the coefficients $\hat{a}_n, \breve{a}_n$ are positive below the last index and vanish there, and $\hat{c}_n, \breve{c}_n > 0$ for $n \geq 1$. Hence the normalized arrays, which describe the same polynomials in the variable $(x - \hat{b}_0)/\hat{a}_0$ (respectively $(x - \breve{b}_0)/\breve{a}_0$), satisfy (2.1). The terminal relation $tR_{2N} = c_{2N}R_{2N-1}$ on the support Σ of the symmetric system, applied once and twice, gives the terminal relations of P on $\{t^2 : t \in \Sigma\}$ and of L on $\{t^2 : t \in \Sigma,\ t \neq 0\}$. These sets have $N+1$ and N points, because Σ is symmetric with $2N+1$ points and contains 0. By Lemma 5.1 they are the supports of the two components.

The identity $tR_{2n} = a_{2n}R_{2n+1} + c_{2n}R_{2n-1}$ gives $P_n = a_{2n}L_n + c_{2n}L_{n-1}$ on the support of L; for $n = N$ it is the terminal relation, with $a_{2N} = 0$. Hence, on $\Sigma \setminus \{0\}$,

$$R_{2a}R_{2b+1} = t\big(a_{2a}L_aL_b + c_{2a}L_{a-1}L_b\big) = \sum_c \big(a_{2a}\breve{g}(a, b; c) + c_{2a}\breve{g}(a-1, b; c)\big)R_{2c+1},$$

and both sides vanish at $t = 0$. Similarly $R_{2a}R_{2b} = (P_aP_b)(t^2) = \sum_c \hat{g}(a,b;c)R_{2c}$ on Σ. Uniqueness of expansions on Σ gives the displayed formulas. The vanishing for $m+n+k$ odd and the relation for two odd indices follow from the symmetry of the measure and from $g(m,n;k)\int R_k^2\,d\mu = \int R_mR_nR_k\,d\mu$, exactly as in the proof of Theorem 7.1. The converse holds because $\hat{g}(a,b;c) = g(2a,2b;2c)$. ∎

Combined with Theorem 2.1, the decomposition gives a parity-split criterion of the same shape as Szwarc's Theorem 3.3: separate conditions on the even-indexed and the odd-indexed coefficients.

Corollary 7.9 *(Parity-split criterion).* *Let $N \geq 1$ and let a symmetric array ($b \equiv 0$) on the indices $0,\dots,2N$ satisfy (2.1). Suppose that*

1. *the odd downward coefficients are constant, $c_{2n+1} = p_0$, with $0 < p_0 \leq 1/2$;*
2. *the even downward coefficients $d_n = c_{2n}$ have positive increments $d_n - d_{n-1}$, $1 \leq n \leq N$, forming a geometric progression with ratio $q \geq 1$.*

Then the array has nonnegative linearization. With $r = p_0/(1-p_0)$ and $N \geq 2$, its normalized even and odd components are (2.4) with parameters (N,q,r) and $(N-1,q,r/q)$, respectively. For $N = 1$ the odd component consists only of $L_0 = 1$, the array is (7.2), and positivity follows directly from Theorem 7.1. If $1/2 < p_0 < 1$ in (i) and (ii) holds, then $g(2,2;2) < 0$.

Proof. For $N = 1$ this is Theorem 7.1. Let $N \geq 2$. The even component has $\hat{a}_n = (1-d_n)(1-p_0)$, $\hat{b}_n = d_n(1-p_0) + (1-d_n)p_0$ and $\hat{c}_n = d_np_0$, so $\hat{a}_0 = 1-p_0$, $\hat{b}_0 = p_0$, and its normalization is $(1-d_n,\,(1-r)d_n,\,rd_n)$. This satisfies Theorem 2.1 with $\beta = (1-r)/r \geq 0$. For the odd component, $\check{a}_n = (1-p_0)(1-d_{n+1})$, $\check{b}_n = (1-p_0)d_{n+1} + p_0(1-d_n)$ and $\check{c}_n = p_0d_n$. Geometric increments give $d_{n+1} - d_1 = qd_n$, so the normalization is

$$\Big(\frac{1-d_{n+1}}{1-d_1},\ \frac{(q-r)d_n}{1-d_1},\ \frac{rd_n}{1-d_1}\Big).$$

Its ratio $\beta = (q-r)/r$ is constant and nonnegative, and its profile $qd_n/(1-d_1)$ has geometric increments with ratio q. Theorem 2.1 applies to both components, and Theorem 7.8 concludes. Lemma 2.2 identifies the parameters. For $p_0 > 1/2$, $g(2,2;2) = \hat{g}(1,1;1)$ is the normalized diagonal coefficient $(1-r)d_1 < 0$ (for $N = 1$ see Theorem 7.1). ∎

With $q = 1$ the corollary is Szwarc's system (7.2), and it recovers Theorem 7.1, including the Krawtchouk polynomials as its even component. For $q > 1$ it gives a three-parameter family of symmetric finite systems with nonnegative linearization whose even components are the dual q-Krawtchouk arrays of Corollary 6.2. Theorem 7.8 also transfers the perturbation criteria: a symmetric array has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Theorem 5.3, or those of Lemma 7.5 around a reference with a margin, provided the set Σ of that lemma contains every coefficient that can be nonzero for the perturbed components, or the remaining coefficients are separately known to stay nonnegative. Lemma 7.5 controls only Σ: for example, starting

from $K_{2,1}$, keeping $b \equiv 0$ and $c_2 = 1$, and setting $c_1 = \frac{1}{2} + \varepsilon$, $a_1 = \frac{1}{2} - \varepsilon$ with $0 < \varepsilon < \frac{1}{2}$ gives $g(2,2;2) = -2\varepsilon/(\frac{1}{2} - \varepsilon) < 0$, although the coefficients positive at the reference remain positive. Conversely, the even component of every array admitted by Theorem 7.6 is a nonsymmetric finite system with nonnegative linearization.

Products of low total degree. Szwarc's inequality method [6, Theorems 3.2 and 3.4] survives the passage to finite systems for all products whose two factors have total degree at most N. Only the terminal coefficients are excluded from its hypotheses. Write a finite system in a positive renormalization $P_n = R_n/\sigma_n$ ($\sigma_0 = 1$) as $xP_n = \gamma_n P_{n+1} + \beta_n P_n + \alpha_n P_{n-1}$, so that $\gamma_n = a_n\sigma_{n+1}/\sigma_n$, $\beta_n = b_n$, $\alpha_n = c_n\sigma_{n-1}/\sigma_n$ and $\gamma_N = 0$.

Theorem 7.10 *(Low products).* *Let a, b, c satisfy (2.1). Suppose that for some positive σ,*

1. *β_n is nondecreasing for $0 \le n \le N-1$, and α_n is nondecreasing for $0 \le n \le N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \le n \le N-2$;*
3. *$\gamma_i \ge \alpha_j$ whenever $0 \le j \le i$ and $i + j \le N - 2$.*

Then $g(n, m; k) \ge 0$ whenever $n + m \le N$. Consequently a product coefficient can be negative only for a triple (n, m, k) in which each two indices have sum greater than N.

Proof. Coefficients in two positive normalizations differ by positive factors, so we may work with the polynomials $\widetilde{P}_n$ of [6, (3.9)], which satisfy $x\widetilde{P}_n = \alpha_{n+1}\widetilde{P}_{n+1} + \beta_n\widetilde{P}_n + \gamma_{n-1}\widetilde{P}_{n-1}$ with $\alpha_{N+1} = 0$. Fix k and let $u(n, m)$ be the coefficient of $\widetilde{P}_k$ in $\widetilde{P}_n\widetilde{P}_m$. Multiplying the product by x in the two ways shows $Hu = 0$ at every point of $\{0, \dots, N\}^2$, where $H = L_1 - L_2$ is the operator of [6, §3.4] built from these coefficients, and $u(n, 0) = \mathbf{1}_{n=k}$, $u(n, m) = u(m, n)$.

We prove $u(n, m) \ge 0$ for $n \ge m$, $n + m \le N$, by induction on m, following [6, Theorem 3.4]. Let v be the indicator of the points (i, j) with $j < m$, $|i - n| < m - j$ and $n + m - i - j$ odd. Every point in the support of v satisfies $j < i$ and $i + j \le n + m - 1 \le N - 1$, so $\langle Hu, v\rangle = 0$ involves only rows where the finite recurrence holds. Hence $0 = \langle u, H^*v\rangle$, and every point in the support of H^*v has $i + j \le n + m \le N$. With $\alpha_0 = 0$ and v set to zero outside its support, transposition gives

$$(H^*v)(i,j) = \alpha_i v(i-1,j) + \gamma_i v(i+1,j) + (\beta_i - \beta_j)v(i,j) - \alpha_j v(i,j-1) - \gamma_j v(i,j+1),$$

as in [6, §3.4]. At (n, m) it is $-\alpha_m < 0$. At the other points it is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j,$$

with $j \le i$. The first occurs only for $i \le N - 1$. The second occurs only at interior points, where $i + j \le n + m - 2 \le N - 2$. The third occurs on the edge $i + j = n + m$. The last occurs on the edge $i - j = n - m$, where $i + j \le N - 2$. Hypotheses (i)–(iii) make all of these nonnegative, and all such points have second coordinate less than m. By the induction hypothesis and $u(i, 0) \ge 0$, $\alpha_m u(n, m) = \sum_{(i,j)\neq(n,m)} (H^*v)(i, j)u(i, j) \ge 0$. For the last statement, $g(n, m; k)$, $g(n, k; m)$ and $g(m, k; n)$ are positive multiples of $\int R_n R_m R_k \, d\mu$. ∎

Corollary 7.11 *(Convex profiles).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1-d_i$ with $0 < r \le 1$ and a convex profile $0 = d_0 < d_1 < \cdots < d_N = 1$, that is, with nondecreasing increments. Then $g(n,m;k) \ge 0$ whenever two of n, m, k have sum at most N. This includes every array (2.4).*

Proof. Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1-d_n)$, so α and $\beta = (1-r)d$ are nondecreasing and $\alpha_n + \gamma_n = \sqrt{r}$ is constant. Convexity and $d_0 = 0$ give $d_i + d_j \le d_{i+j}$, so $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \ge \sqrt{r}\,(1 - d_{i+j}) > 0$ for $i + j \le N-2$. Theorem 7.10 applies. ∎

For such profiles, positivity can therefore fail only on triples with all pairwise sums above N. The convex counterexamples recorded in Section 10 fail at (N, N, N), which is such a triple.

The cone argument extends to all products if the monotonicity conditions in (i)–(ii) are imposed through the terminal index N and the edge inequalities in (iii) are imposed for every $0 \le j \le i \le N-1$.

Theorem 7.12 *(All products).* *Let a, b, c satisfy (2.1), and put $\gamma_N = 0$. Suppose that for some positive σ,*

1. *β_n and α_n are nondecreasing for $0 \le n \le N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \le n \le N$;*
3. *$\gamma_i \ge \alpha_j$ whenever $0 \le j \le i \le N-1$.*

Then the system has nonnegative linearization.

Proof. Use the notation of the proof of Theorem 7.10. Since $\alpha_{N+1} = 0$, the relation $x\widetilde{P}_N = \beta_N \widetilde{P}_N + \gamma_{N-1}\widetilde{P}_{N-1}$ is the terminal relation, which holds on the support. So $Hu = 0$ at every point of $\{0, \dots, N\}^2$, as noted there. We prove $u(n,m) \ge 0$ for all $n \ge m$ by induction on m. Let v be the indicator of the points (i,j) with $0 \le i \le N$, $j < m$, $|i-n| < m-j$ and $n+m-i-j$ odd. Then $0 = \langle Hu, v\rangle = \langle u, H^*v\rangle$, and H^*v is given by the formula in that proof. At $i = N$ the term $\gamma_N v(N+1, j)$ is absent; this is the convention $\gamma_N = 0$. Hence $H^*v(n,m) = -\alpha_m$, and at every other point (i,j) with $H^*v(i,j) \ne 0$ we have $j < m$, $j \le i \le N$, and $H^*v(i,j)$ is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j.$$

The last one occurs only on the edge $i - j = n - m$, where $i \le n - 1 \le N - 1$. Hypotheses (i)–(iii) make all of these nonnegative. The induction hypothesis then gives $\alpha_m u(n,m) \ge 0$, and the case $n < m$ follows by symmetry. ∎

Corollary 7.13 *(Heavy top step).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1-d_i$ with $0 < r \le 1$ and an increasing profile $0 = d_0 < d_1 < \cdots < d_N = 1$ with $d_{N-1} \le \frac{1}{2}$. Then the system has nonnegative linearization.*

Proof. Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1 - d_n)$ for $n < N$, $\gamma_N = 0$, and $\beta_n = (1-r)d_n$. So α and β are nondecreasing, and $\alpha_n + \gamma_n = \sqrt{r}$ for every $0 \le n \le N$, because $d_N = 1$. For $j \le i \le N-1$, $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \ge \sqrt{r}\,(1 - 2d_{N-1}) \ge 0$. Theorem 7.12 applies. ∎

The profile need not be convex. The corollary contains the arrays (2.4) with $q \ge 2$, since then $d_{N-1} = (q^{N-1}-1)/(q^N-1) \le \frac{1}{2}$, but not the Krawtchouk arrays with $N \ge 3$. The bound $d_{N-1} \le \frac{1}{2}$ cannot be relaxed in general: for $N = 2$ and $r = 1$, $g(2,2;2) = (1-2d_1)/(1-d_1)$. For constant-ratio arrays with a convex profile, $d_i + d_j \le d_{i+j} \le 1$ whenever $i + j \le N$, so the only terms that can be negative in the proof are the edge terms $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j)$ with $i + j > N$. For the Krawtchouk arrays these edge coefficients are $\sqrt{r}\,(N - i - j)/N$; any negative weighted contributions they make to the cone identity must be compensated by its other terms.

Deep triples and reversal. Call (n, m, k) deep if each two of its indices have sum greater than N. The reversed array $a'_i = c_{N-i}$, $b'_i = b_{N-i}$, $c'_i = a_{N-i}$ need not satisfy $a'_0 = 1$, $b'_0 = 0$. Its polynomials R'_i are defined by the reversed recurrence from $R'_0 = 1$, so $R'_1 = (x - b_N)/c_N$, and they still satisfy $R'_i(1) = 1$; its matrix is $\Pi J \Pi$, where Π reverses the order of the indices. Write g' for its product coefficients.

Lemma 7.14 *(Reversal identity).* *On the support, $R_N R'_t = R_{N-t}$ for $0 \le t \le N$.*

Proof. From $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$ for the array and for its reversal, $e_N^{\mathsf{T}} R'_t(J) = e_{N-t}^{\mathsf{T}}$. Hence $e_0^{\mathsf{T}} R_N(J) R'_t(J) = e_N^{\mathsf{T}} R'_t(J) = e_0^{\mathsf{T}} R_{N-t}(J)$. Both matrices commute with J and e_0 is cyclic, so $R_N(J) R'_t(J) = R_{N-t}(J)$, which is the claim on the spectrum. ∎

Theorem 7.15 *(Deep reduction).* *For every array satisfying (2.1) and every deep triple (x, y, z),*

$$g(x, y; z) = \sum_{t=0}^{N} g'(N - x, N - y; t)\, g(N, N - t; z).$$

Moreover $g(N, N; N - l)$ is the coefficient of R'_l in R_N. Hence all coefficients are nonnegative if (a) $g \ge 0$ on non-deep triples, (b) $g'(i, j; t) \ge 0$ whenever $i + j < N$, and (c) R_N is a nonnegative combination of $R'_0, \dots, R'_N$.

Proof. By Lemma 7.14, $R_x = R_N R'_{N-x}$ and $R_y = R_N R'_{N-y}$ on the support. Expanding $R'_{N-x} R'_{N-y} = \sum_t g'(N - x, N - y; t) R'_t$ and using Lemma 7.14 again gives $R_x R_y = \sum_t g'(N - x, N - y; t) R_N R_{N-t}$. Expanding $R_N R_{N-t}$ gives the formula. Writing $R_N = \sum_l e_l R'_l$, Lemma 7.14 gives $R_N^2 = \sum_l e_l R_{N-l}$, so $e_l = g(N, N; N - l)$. For the last statement, $(N - x) + (N - y) < N$, so (b) makes the weights g' nonnegative. It remains to see $g(N, n; z) \ge 0$. If (N, n, z) is not deep this is (a). Otherwise $n + z > N$, and $g(N, n; z)$ is a positive multiple of $g(n, z; N) = \sum_t g'(N - n, N - z; t)\, g(N, N - t; N)$; here $g(N, N - t; N)$ is a positive multiple of $g(N, N; N - t)$, which is nonnegative by (c). ∎

Theorem 7.16 *(Palindromic criterion).* *Let a, b, c satisfy (2.1) with $a_n = c_{N-n}$ and $b_n = b_{N-n}$ for all n, and satisfy the hypotheses of Theorem 7.10 for some positive σ. Then the system has nonnegative linearization.*

Proof. The reversed array equals the array, so $R' = R$ and $g' = g$. Theorem 7.10 gives $g \geq 0$ on all non-deep triples, in particular $g'(i, j; t) \geq 0$ for $i + j < N$. Since $R_N = R'_N$ is a single reversed polynomial, $g(N, N; \cdot) \geq 0$. Theorem 7.15 gives the deep triples. ∎

Corollary 7.17 *(Monotone palindromic systems).* *Let $b \equiv 0$, $a_n = 1 - c_n$, and let $0 = c_0 < c_1 \leq c_2 \leq \cdots \leq c_N = 1$ satisfy $c_n + c_{N-n} = 1$. Then the system has nonnegative linearization. This includes the symmetric Krawtchouk array $c_n = n/N$.*

Proof. The array is palindromic. With $\sigma \equiv 1$, $\alpha_n = c_n$ is nondecreasing, $\beta \equiv 0$ and $\alpha_n + \gamma_n = 1$. If $j \leq i$ and $i + j \leq N - 2$, then $j < N - i$, so $\alpha_j = c_j \leq c_{N-i} = 1 - c_i = \gamma_i$. Theorem 7.16 applies. ∎

Within the palindromic subclass $b \equiv 0$, $c_n + c_{N-n} = 1$, the hypotheses of Corollary 7.17 are monotonicity inequalities. They contain the symmetric Krawtchouk endpoint, where the perturbation criterion of Section 5 gives no neighbourhood and where the convex counterexamples of Section 10 show that convexity alone does not suffice.

Combined with Theorem 7.8, Corollary 7.17 gives an inequality criterion for symmetric systems: an array with $b \equiv 0$ on $0, \ldots, 2N$ has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Corollary 7.17. For Szwarc's system (7.2) at $p_0 = 1/2$ the components are the symmetric Krawtchouk arrays on $N + 1$ and N points (Corollary 7.9 with $q = r = 1$), so this case is covered by monotonicity inequalities within the palindromic subclass.

The reversal route cannot reach systems whose diagonal increases at the top once $N \geq 3$. If $b_{N-1} < b_N$, the reversed array has $g'(1, 1; 1) = (b'_1 - b'_0)/a'_0 = (b_{N-1} - b_N)/c_N < 0$, and condition (b) of Theorem 7.15 requires $g'(1, 1; 1) \geq 0$ when $1 + 1 < N$. In particular it fails for the Krawtchouk arrays with $r < 1$ and $N \geq 3$. Szwarc's array (7.2) has $b \equiv 0$, so this argument does not apply to it. Its reversal replaces p_0 by $1 - p_0$ and gives $g'(2, 2; 2) = (2p_0 - 1)/(Np_0) < 0$ for $p_0 < 1/2$, which violates condition (b) for the array of length $2N + 1$ when $4 < 2N$, that is, $N \geq 3$. Miclo's palindromic criterion [5, Theorem 31] requires $2^n P(n, n + 1)$ to be nonincreasing on $[0, N/2]$, which the symmetric Krawtchouk chain violates for $N > 2$.

8. Log-concave profiles

Theorem 2.1 requires the increments of the profile $d_i = b_i + c_i$ to be geometric. This section replaces that equality by inequalities on the Newton coefficients of the profile, for arrays with constant ratio b_i/c_i. Write

$$\nu_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} d_i \quad (1 \leq k \leq N), \qquad d_n = \sum_{k=1}^{n} \binom{n}{k} \nu_k \quad (0 \leq n \leq N);$$

below we put $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$. We say that $(\nu_k)_{1\le k\le N}$ is *log-concave without internal zeros* if all $\nu_k \ge 0$, the indices with $\nu_k > 0$ form an interval $\{1,\dots,K\}$, and $\nu_k^2 \ge \nu_{k-1}\nu_{k+1}$ for $2 \le k \le N-1$. Note $\nu_1 = d_1 > 0$.

Theorem 8.1 *(Log-concave Newton criterion).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$, and put $a_i = 1 - d_i$, $b_i = (1-r)d_i$, $c_i = rd_i$. Then the system has nonnegative linearization.*

The hypotheses are inequalities in the profile. They hold for the Krawtchouk arrays ($\nu = (1/N, 0, \dots, 0)$), for all arrays of Theorem 2.1 (ν_k proportional to $(q-1)^{k-1}$, log-concave with equality). The hypotheses also hold for every admissible convex quadratic profile: $\nu_1 > 0$, $\nu_2 \ge 0$ when $N \ge 2$, and $\nu_k = 0$ for $3 \le k \le N$. The hypotheses exclude the known counterexamples with constant ratio. The convex counterexamples recorded in Section 10 have Newton coefficients of both signs. For the combinations $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ discussed there, with $q > 1$, $N \ge 3$ and $0 < \lambda < 1$, one has $\nu_1 = (1-\lambda)/N + \lambda/[N]_q$ and $\nu_k = \lambda(q-1)^{k-1}/[N]_q$ for $2 \le k \le N$, so $\nu_2^2 - \nu_1\nu_3 = -\lambda(1-\lambda)(q-1)^2/(N[N]_q) < 0$.

A rescaled system and a kernel. Fix r and put $t = (1-r)/\sqrt{r} \ge 0$. If p_i is defined by $p_0 = 1$ and $yp_i = a_ip_{i+1} + td_ip_i + d_ip_{i-1}$ $(0 \le i < N)$, then $R_i(x) = r^{i/2}p_i(x/\sqrt{r})$, so the coefficients H^k_{mn} in $p_mp_n = \sum_k H^k_{mn}p_k$ on the support satisfy $g(m,n;k) = r^{(m+n-k)/2}H^k_{mn}$. It suffices to show $H^k_{mn} \ge 0$. Put

$$e_i = \prod_{j=0}^{i-1} \frac{a_j}{d_{j+1}}, \qquad E_i = e_ip_i.$$

Then $yE_i = d_{i+1}E_{i+1} + td_iE_i + a_{i-1}E_{i-1}$ on the support for $0 \le i \le N$, with the terms $d_{N+1}E_{N+1}$ and $a_{-1}E_{-1}$ omitted; at $i = N$ this is the terminal relation. Write $E_mE_n = \sum_k L^k_{mn}E_k$, so that $H^k_{mn} = L^k_{mn}e_k/(e_me_n)$, and

$$K_k(z,w) = \sum_{m,n=0}^{N} L^k_{mn}z^mw^n.$$

Let $\theta_z = z\,\partial_z$ and let $d(\theta) = \sum_k \nu_k\binom{\theta}{k}$ act on polynomials by $d(\theta_z)z^i = d_iz^i$. For $F(z) = \sum_{i=0}^N E_iz^i$ the recurrence reads $yF(z) = \mathcal{L}_zF(z)$ with

$$\mathcal{L}_z = (z^{-1} + t - z)\,d(\theta_z) + z,$$

because $d_0 = 0$ removes the term $z^{-1}E_0$ and $1 - d_N = 0$ removes the term $z^{N+1}E_N$. Applying this to $F(z)F(w) = \sum_k K_k(z,w)E_k$ in the two variables separately and comparing coefficients of the basis E_k gives $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$. Since K_k is symmetric, $K_k = \sum_{\ell,a\ge0}\omega_{\ell,a}S^aZ^\ell$ with $S = z + w$, $Z = zw$, and $K_k(z,0) = z^k$ gives $\omega_{0,a} = \mathbf{1}_{a=k}$. As K_k has degree at most N in z, comparing the coefficients of $z^{\ell+a}w^\ell$ by induction on ℓ gives $\omega_{\ell,a} = 0$ whenever $\ell + a > N$.

The kernel recurrence. Put $f_n(\ell) = \sum_{h\ge 0}\binom{\ell}{h}\nu_{n+h}$ with $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$, so $f_0(\ell) = d_\ell$, and

$$W_{p,q}(\ell,a) = \sum_{n\ge p}\binom{n-p+q}{q}\binom{a}{n} f_n(\ell) \ge 0,$$

and $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$.

Lemma 8.2. *For all $\ell, a \ge 0$,*

$$\begin{aligned} d_{\ell+1}\omega_{\ell+1,a} = a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\ge 0}(-1)^j\Big(&t\,A_j(\ell-j,a+2j+1)\,\omega_{\ell-j,a+2j+1}\\ &+ B_j(\ell-j,a+2j+2)\,\omega_{\ell-j,a+2j+2}\\ &+ C_j(\ell-j-1,a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned} \tag{8.1}$$

where $a_i = 1 - d_i$ is extended by the polynomial d_i for $i > N$ and terms with a negative index are omitted.

Proof. Since $S^aZ^\ell = \sum_i\binom{a}{i}z^{\ell+i}w^{\ell+a-i}$, Vandermonde's identity $\binom{\ell+i}{k} = \sum_{p+q=k}\binom{\ell}{q}\binom{i}{p}$ and $\binom{a}{i}\binom{i}{p} = \binom{a}{p}\binom{a-p}{i-p}$ give

$$d(\theta_z)\,S^aZ^\ell = \sum_{p,q\ge 0}\nu_{p+q}\binom{\ell}{q}\binom{a}{p}z^pS^{a-p}Z^\ell.$$

Hence $(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell$ equals $(z-w)S^aZ^\ell$ plus $\sum_{p,q}\nu_{p+q}\binom{\ell}{q}\binom{a}{p}S^{a-p}Z^\ell$ times $(z^{p-1} - w^{p-1}) + t(z^p - w^p) - (z^{p+1} - w^{p+1})$. Divide by $z-w$, using $\chi_n = (z^{n+1} - w^{n+1})/(z-w) = \sum_{0\le j\le\lfloor n/2\rfloor}(-1)^j\binom{n-j}{j}S^{n-2j}Z^j$ for $n \ge 0$, $\chi_{-1} = 0$ and $(z^{-1} - w^{-1})/(z-w) = -Z^{-1}$. Since $\sum_q\binom{\ell}{q}\nu_{p+q} = f_p(\ell)$, the quotient is

$$\begin{aligned} &- d_\ell S^aZ^{\ell-1} + a_{\ell+a}S^aZ^\ell\\ &+ \sum_{j\ge 0}(-1)^j\Big(tA_j(\ell,a)S^{a-2j-1}Z^{\ell+j} + B_j(\ell,a)S^{a-2j-2}Z^{\ell+j} + C_j(\ell,a)S^{a-2j-2}Z^{\ell+j+1}\Big). \end{aligned}$$

Here $-d_\ell S^aZ^{\ell-1}$ comes from $p=0$; the term $a_{\ell+a}S^aZ^\ell$ combines S^aZ^ℓ with the parts $j=0$ of $-\chi_p$, which sum to $-d_{\ell+a}S^aZ^\ell$ by Vandermonde; and the three families from χ_{p-2}, $t\chi_{p-1}$ and $-(\chi_p - S^p)$, whose coefficients $\binom{p-2-j}{j}$, $\binom{p-1-j}{j}$ and $\binom{p-1-j}{j+1}$ are those of B_j, A_j and C_j. Summing over ℓ, a with weights $\omega_{\ell,a}$ and taking the coefficient of S^aZ^ℓ gives (8.1), because the monomials S^aZ^ℓ are linearly independent. ∎

The log-concavity inequality.

Lemma 8.3. *Under the hypotheses of Theorem 8.1, for $i, l \ge 1$ and $\ell \ge 0$,*

$$f_i(\ell)\,f_l(\ell+1) \ge d_{\ell+1}\,f_{i+l}(\ell).$$

Proof. For a nonnegative finitely supported sequence u that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_iu_{i+2}) + (u_iu_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(0, \nu_1, \nu_2, \dots)$ has this property, and $f_i(\ell)$ is obtained from it by applying $u \mapsto v$ ℓ times. For such a sequence g, balanced products dominate: $g_ig_l \geq g_0g_{i+l}$ and $g_ig_{l+1} \geq g_1g_{i+l}$ for $i, l \geq 1$. Adding these with $g = f(\ell)$ and using $g_l + g_{l+1} = f_l(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1) = d_{\ell+1}$ gives the claim. ∎

Lemma 8.4. *Under the hypotheses of Theorem 8.1, for all $j \geq 1$ and $\ell, a \geq 0$,*

$$\begin{aligned} A_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}(\ell+1,a-2) &\geq 4d_{\ell+1}A_j(\ell,a), \\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\geq 4d_{\ell+1}B_j(\ell,a), \\ B_0(\ell,a)C_{j-1}(\ell+1,a-2) &\geq 4d_{\ell+1}C_j(\ell,a), \end{aligned}$$

with $W_{p,q}(\ell,a) = 0$ for $a < 0$.

Proof. Use $A_0(\ell,a) \geq af_1(\ell)$ and $B_0(\ell,a) \geq \binom{a}{2}f_2(\ell)$, expand the second factor, apply Lemma 8.3 to $f_1(\ell)f_n(\ell+1)$ and $f_2(\ell)f_n(\ell+1)$, and use $a\binom{a-1}{n} = (n+1)\binom{a}{n+1}$ and $\binom{a}{2}\binom{a-2}{n} = \binom{n+2}{2}\binom{a}{n+2}$. After division by $d_{\ell+1}$ the three left sides are bounded below by $\sum_m \gamma_m \binom{a}{m} f_m(\ell)$ with

$$\gamma_m = \binom{m+1}{2}\binom{m-j-2}{j-1}, \qquad \binom{m}{2}\binom{m-j-3}{j-1}, \qquad \binom{m}{2}\binom{m-j-2}{j},$$

while the right sides are $4\sum_m \gamma'_m \binom{a}{m} f_m(\ell)$ with $\gamma'_m = \binom{m-j-1}{j}$, $\binom{m-j-2}{j}$ and $\binom{m-j-1}{j+1}$. Where $\gamma'_m > 0$ the quotients are

$$\frac{jm(m+1)}{2(m-j-1)}, \qquad \frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)},$$

on $m \geq 2j+1$, $m \geq 2j+2$ and $m \geq 2j+2$ respectively. They increase with j for fixed m; at $j = 1$ their minima over integers are 5, 5 and 6. All quotients exceed 4, and $f_m(\ell) \geq 0$. ∎

Proof of Theorem 8.1. Let $\Omega = \{(\ell,a) : \ell, a \geq 0,\ \ell + a \leq N\}$, on which ω is supported. Read (8.1) as moves into the node $(\ell+1, a)$, each divided by $d_{\ell+1} > 0$: the straight move from (ℓ, a) with weight $a_{\ell+a} \geq 0$ on Ω, and for each j the three moves from $(\ell - j, a + 2j + 1)$, $(\ell - j, a + 2j + 2)$, $(\ell - j - 1, a + 2j + 2)$ with weights tA_j, B_j, C_j. Every move raises ℓ. Let $\mathcal{U}$ be the straight moves, $\mathcal{P}$ the moves with j even and $\mathcal{Q}$ the moves with j odd, all with nonnegative weights, so that $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\beta = \omega|_{\ell=0}$.

We claim $\mathcal{P}^2 \geq 4\mathcal{Q}$ entrywise. A move of $\mathcal{Q}$ with index j from $x = (\ell, a)$ is matched by the two-step paths in $\mathcal{P}^2$ that start with A_0 or B_0 from x (index 0, landing at $(\ell+1, a-1)$ or $(\ell+1, a-2)$, which lie in Ω) and continue with the move of the same type and index $j-1$ (even), as in Lemma 8.4: A_j by A_0 then B_{j-1} and B_0 then A_{j-1}; B_j by B_0 then B_{j-1}; C_j by B_0 then C_{j-1}. These paths end at the target of the matched move, carry the same power of t, and have the extra denominator $d_{\ell+1}$ of the intermediate node. Distinct

moves of $\mathcal{Q}$ have distinct displacements, so their paths are distinct, and Lemma 8.4 gives the claim.

Now show by induction on the level that $\omega \ge 0$ and $\omega \ge \frac{1}{2}\mathcal{P}\omega$. Both hold at $\ell = 0$. If they hold below a node y at level $\ell + 1 \ge 1$, then, all weights being nonnegative, $(\mathcal{P}\omega)(y) \ge \frac{1}{2}(\mathcal{P}^2\omega)(y) \ge 2(\mathcal{Q}\omega)(y)$, and

$$\omega(y) = (\mathcal{U}\omega)(y) + (\mathcal{P}\omega)(y) - (\mathcal{Q}\omega)(y) \ge \tfrac{1}{2}(\mathcal{P}\omega)(y) \ge 0.$$

Hence all $\omega_{\ell,a} \ge 0$, so K_k has nonnegative coefficients in z, w, so $L^k_{mn} \ge 0$, $H^k_{mn} \ge 0$ and $g(m, n; k) \ge 0$. ∎

Remark 8.5. The argument is valid coefficientwise in t. With independent parameters in $yp_i = a_i p_{i+1} + u d_i p_i + v d_i p_{i-1}$, weighted homogeneity ($\deg y = \deg u = 1$, $\deg v = 2$) then shows that each linearization coefficient is a polynomial in (u, v) with nonnegative coefficients: for $D = m + n - k \ge 0$ it has the form $\sum_{0 \le j \le \lfloor D/2 \rfloor} h_j u^{D-2j} v^j$ with $h_j \ge 0$, and it vanishes for $D < 0$. Consequently every $g(m, n; k)$ of the array of Theorem 8.1, as a function of r, equals $\sum_j h_j r^j (1 - r)^{D-2j}$; multiplying the j-th term by $(r + (1 - r))^j = 1$ shows that its coefficients in the Bernstein basis of degree D are $\binom{D}{s}^{-1} \sum_{j \le s \le 2j} h_j \binom{j}{s-j} \ge 0$. The diagnostics `lc_kernel_check.py` verify Lemmas 8.2–8.4 and the matching $\mathcal{P}^2 \ge 4\mathcal{Q}$ in exact arithmetic on random log-concave profiles.

9. Limits of inequality criteria

Theorems 7.10 and 7.16 control every triple except the deep ones by inequalities, Theorem 8.1 controls all triples of constant-ratio arrays with log-concave Newton profiles, and Section 10 records shape conditions that fail on deep triples. This section shows that part of the failure is structural near the symmetric endpoint. For uniformly sampled polynomial deformation directions, and for the analytic and boundary-layer extensions specified below, admission at every sufficiently large symmetric Krawtchouk system imposes two first-order boundary equalities. This does not exclude arbitrary uniform inequality criteria, nor families of admissible arrays that are not of these forms. Write $K_{N,r}$ for the Krawtchouk array (2.4) with $q = 1$, and use the coordinates $b_1, \dots, b_{N-1}, c_1, \dots, c_N$ of a stochastic array with $b_0 = 0$, $a_0 = 1$, $a_N = 0$, $b_N = 1 - c_N$ and $a_n = 1 - b_n - c_n$ otherwise. An affine change of x, together with the positive normalization $R_n(x_0) = 1$ at the largest support point x_0, reduces every finite system to this form. A direction $v = (\delta b_1, \dots, \delta b_{N-1}, \delta c_1, \dots, \delta c_N)$ changes these coordinates; we put $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$.

Lemma 9.1 *(Corner value).* *For $0 < r \le 1$, the array $K_{N,r}$ has $g(N, N; N) = (1 - r)^N$.*

Proof. In the model of Lemma 5.2, $U_N = y_1 \cdots y_N$ is the only normalized basis element containing $y_{\{1,\dots,N\}}$, and $U_N^2 = \prod_i \big((1 - r)y_i + r\big)$ has coefficient $(1 - r)^N$ at that monomial. ∎

Thus the corner coefficient vanishes at $r = 1$, where it is a structural zero, together with the other perimeter and parity zeros of the palindromic array. The next lemma computes its first-order change exactly.

Lemma 9.2 *(Corner identity).* *Let $N \geq 2$ and $s_N = (-1)^{\lfloor N/2 \rfloor + 1} \binom{N-1}{\lfloor (N-1)/2 \rfloor}^{-1}$. Put $F = \delta b + 2\delta c$ if N is even and $F = \delta b$ if N is odd. Then*

$$\frac{d}{d\varepsilon}\, g(N, N; N)\Big|_{K_{N,1}+\varepsilon v,\ \varepsilon=0} \;=\; s_N \sum_{n=0}^{N} (-1)^n \binom{N}{n} F(n).$$

Proof. At $K_{N,1}$ the nodes are $\lambda_j = 1 - 2j/N$, with masses $w_j = 2^{-N}\binom{N}{j}$ and $\pi_n = \binom{N}{n}$ in Lemma 5.1. Moreover $\binom{N}{n} R_n(\lambda_j)$ is the binary Krawtchouk polynomial, so that

$$\sum_{j=0}^{N} w_j R_n(\lambda_j) s^j \;=\; 2^{-N}(1+s)^{N-n}(1-s)^n. \tag{9.1}$$

The vectors $u_j = (R_n(\lambda_j))_n$ and $v_j = (w_j \pi_n R_n(\lambda_j))_n$ are right and left eigenvectors of J with $v_i^{\mathsf{T}} u_j = \mathbf{1}_{i=j}$. Since $R_N(\lambda_j) = (-1)^j$, the matrix $\Pi = R_N(J)$ reverses the indices. Let $J_\varepsilon = J + \varepsilon E$ be the perturbed recurrence matrix and R_N^ε the perturbed polynomial. The matrix $R_N^\varepsilon(J_\varepsilon)$ commutes with J_ε and has first row e_N^{T}, by (5.1). Its derivative X at $\varepsilon = 0$ therefore satisfies $[J, X] = [\Pi, E]$ and $e_0^{\mathsf{T}} X = 0$. In the eigenbasis, $\widehat{X}_{ij} = v_i^{\mathsf{T}} X u_j$ equals $((-1)^i - (-1)^j)\widehat{E}_{ij}/(\lambda_i - \lambda_j)$ for $i \neq j$, where $\widehat{E}_{ij} = v_i^{\mathsf{T}} E u_j$, and $\sum_i \widehat{X}_{ij} = 0$. By (5.1), and since $\pi_N = 1$, the derivative in the lemma is $h_N := e_N^{\mathsf{T}} X e_N = \sum_{i,j} (-1)^{i+j} \widehat{X}_{ij} w_j$. Eliminating $\widehat{X}_{jj}$ gives

$$h_N \;=\; -2 \sum_{i+j \text{ odd}} \widehat{X}_{ij} w_j \;=\; N \sum_{i \neq j} \widehat{E}_{ij} w_j \frac{(-1)^i - (-1)^j}{i - j}.$$

Perturbing row n alone means $E = e_n \rho^{\mathsf{T}}$ and $\widehat{E}_{ij} = w_i \pi_n R_n(\lambda_i)\, \rho^{\mathsf{T}} u_j$. Write i for the imaginary unit and insert

$$\frac{(-1)^i - (-1)^j}{i - j} \;=\; -\pi \int_0^1 \sin \pi\big(\theta i + (1-\theta) j\big)\, d\theta,$$

which also vanishes for $i = j$. Then

$$h_N \;=\; -\pi N \pi_n \int_0^1 \operatorname{Im}\big(A(\theta) B(\theta)\big)\, d\theta,$$
$$A \;=\; \sum_i w_i R_n(\lambda_i) e^{\mathrm{i}\pi\theta i}, \qquad B \;=\; \sum_j w_j (\rho^{\mathsf{T}} u_j) e^{\mathrm{i}\pi(1-\theta) j},$$

and both sums are given by (9.1). Put $\phi = \pi\theta/2$.

- Coordinate b_n, $1 \leq n < N$, so $\rho = e_n - e_{n+1}$. Then

  $$AB \;=\; -(-1)^n\, \mathrm{i}^{N-1} \cos^N \phi\, \sin^{N-1} \phi\, e^{-\mathrm{i}\phi},$$

  and $h_N = s_N (-1)^n \binom{N}{n}$. The integral reduces to $\int_0^{\pi/2} \cos^{N+1}\phi\, \sin^{N-1}\phi\, d\phi$ (N even) or $\int_0^{\pi/2} \cos^N \phi\, \sin^N \phi\, d\phi$ (N odd), and each equals $|s_N|/(2N)$.

- Coordinate c_n, $1 \le n < N$, so $\rho = e_{n-1} - e_{n+1}$. Now B acquires the factor $e^{\mathrm{i}\phi}/\cos\phi$, and the integrand carries $\sin((N-1)\pi/2)$. The contribution vanishes for odd N. For even N it is twice that of b_n, since the two beta integrals have ratio 2.

- Coordinate c_N, so $\rho = e_{N-1} - e_N$ in row N. The same computation with $n = N$ gives s_N.

Summing these contributions with the conventions $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$ gives the stated alternating sum. ∎

Definition 9.3. A *uniform rule* is a pair of real functions β, γ on $[0,1]$. It defines, for each N, the direction v_N with $\delta b_n = \beta(n/N)$ for $1 \le n < N$ and $\delta c_n = \gamma(n/N)$ for $1 \le n \le N$. A set of arrays *admits* the rule at the symmetric endpoint if there is N_0 such that for every $N \ge N_0$ there is $\varepsilon_N > 0$ with $K_{N,1} + \varepsilon v_N$ in the set for all $0 < \varepsilon < \varepsilon_N$. Put

$$L_{\mathrm{even}} = \beta(0) + 2\gamma(0) + \beta(1) + \gamma(1), \qquad L_{\mathrm{odd}} = \beta(0) - \beta(1) - \gamma(1).$$

Theorem 9.4 *(Forced equalities for uniform criteria).* *Let β, γ be polynomials. If a set of arrays with property (P) admits the rule (β, γ) at the symmetric endpoint, then $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$. If $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$, the first-order change of $g(N, N; N)$ along v_N vanishes for every N above the degrees of β and γ.*

Proof. For a polynomial p of degree less than N, $\sum_{n=0}^{N} (-1)^n \binom{N}{n} p(n/N) = 0$. Only boundary terms survive in Lemma 9.2: the derivative equals $-s_N L_{\mathrm{even}}$ for even N and $-s_N L_{\mathrm{odd}}$ for odd N. The sign of s_N changes from N to $N+2$. If, for instance, $L_{\mathrm{even}} \ne 0$, then the derivative is negative for every large even N in one residue class modulo 4. For such N, $g(N, N; N) < 0$ at $K_{N,1} + \varepsilon v_N$ for all small $\varepsilon > 0$, since the coefficient vanishes at $\varepsilon = 0$. ∎

The known positive deformations satisfy both equalities: the Krawtchouk direction $\beta(x) = x$, $\gamma(x) = -x$, palindromic deformations with $\beta = 0$, γ odd about $\frac{1}{2}$ and $\gamma(0) = \gamma(1) = 0$ (Corollary 7.17), and the tangent direction $\beta = 0$, $\gamma(x) = x(x-1)$ of the geometric family at $q = 1$ (Theorem 2.1). The two functionals are independent already on constant rules, where $L_{\mathrm{even}} = 2\beta + 3\gamma$ and $L_{\mathrm{odd}} = -\gamma$. Consequently, for every $d \ge 0$, no set of arrays with property (P) admits all rules of an open subset of the polynomial rules of degree at most d. In this sense, a criterion that admits the symmetric Krawtchouk systems through polynomial uniform rules must impose the two equalities on them. The geometric criterion and Corollary 7.17 are of this mixed type.

The polynomial hypothesis can be replaced by analyticity on a fixed region. Let $U(x) = \int_0^1 \log|x - t|\, dt$ and $\Omega = \{x \in \mathbb{C} : U(x) \le -1\}$, the region of convergence of equispaced interpolation. It contains $[0,1]$, has nonanalytic boundary points (with vertical tangents) at 0 and 1, and meets the line $\operatorname{Re} x = \frac{1}{2}$ at $\operatorname{Im} x = \pm 0.26276\ldots$.

Lemma 9.5 *(Nörlund–Rice bound).* *If f is analytic on an open set containing Ω, then $\sum_{n=0}^{N} (-1)^n \binom{N}{n} f(n/N) = O(\sqrt{N}\, e^{-\eta N})$ for some $\eta > 0$.*

Proof. The residues of $1/\prod_{k=0}^{N}(x-k/N)$ give

$$\sum_{n=0}^{N}(-1)^n\binom{N}{n}f(n/N) \;=\; (-1)^N\frac{N!}{N^N}\,\frac{1}{2\pi i}\oint_\Gamma \frac{f(x)\,dx}{\prod_{k=0}^{N}(x-k/N)}$$

for any cycle Γ in the domain of f that winds once around $[0,1]$. Since U is continuous and tends to ∞, the compact sets $\{U\le -1+t\}$ shrink to Ω as $t\downarrow 0$. Choose $\eta>0$ with $\{U\le -1+3\eta\}$ inside the domain of f, and a bounded polygonal open set V with $\{U\le -1+\eta\}\subset V$ and $\overline{V}\subset\{U<-1+3\eta\}$. The oriented boundary Γ of V, possibly with several components, is a cycle outside $\{U\le -1+\eta\}$ that winds once around every node k/N and bounds a region inside the domain of f. With $d=\operatorname{dist}(\Gamma,[0,1])>0$, the derivative of $t\mapsto\log|x-t|$ on $[0,1]$ is bounded by $1/d$ for $x\in\Gamma$, so a Riemann-sum estimate gives $\sum_{k=0}^{N}\log|x-k/N|\ge NU(x)-C_\Gamma\ge N(-1+\eta)-C_\Gamma$ with C_Γ independent of N. Stirling's bound $N!/N^N\le e\sqrt{N}e^{-N}$ completes the proof. ∎

Corollary 9.6. *Let β,γ be analytic on an open set containing Ω. If a set of arrays with property (P) admits the rule (β,γ) at the symmetric endpoint, then $L_{\mathrm{even}}=L_{\mathrm{odd}}=0$. When these equalities hold, the first-order change of $g(N,N;N)$ along v_N is $O(|s_N|\sqrt{N}\,e^{-\eta N})$; it need not vanish.*

Proof. By Lemma 9.5, the alternating sum of $\beta+2\gamma$ or β is $O(\sqrt{N}e^{-\eta N})$. The derivative in Lemma 9.2 is therefore $s_N(-L_{\mathrm{even}}+o(1))$ for even N and $s_N(-L_{\mathrm{odd}}+o(1))$ for odd N, and the argument of Theorem 9.4 gives the equalities. The rule $\beta(x)=e^x-1-(e-1)x$, $\gamma=0$ has $L_{\mathrm{even}}=L_{\mathrm{odd}}=0$ and derivative $s_N(1-e^{1/N})^N\neq 0$. ∎

Deformations may also depend on N in other ways. For each fixed N the symmetric endpoint is a limit of systems whose product coefficients are strictly positive on the whole triangle; the corner nevertheless constrains families that are regular in N. The first statement rests on the following lemma, valid for every finite system. Write μ for the orthogonality measure and x_0 for its largest support point, so that $R_n(x_0)=1$.

Lemma 9.7 *(Lowering the top mass).* *Let $0\le\varepsilon<\mu(\{x_0\})$, let $\mu_\varepsilon=\mu-\varepsilon\delta_{x_0}$, and let R_n^ε be its polynomials, normalized at x_0. Put $\pi_n^\varepsilon=\bigl(\int (R_n^\varepsilon)^2\,d\mu_\varepsilon\bigr)^{-1}$ and $\pi_n=\pi_n^0$; these inverse norms are not rescaled to impose $\pi_0^\varepsilon=1$. The recurrence coefficients in (1) are taken in the fixed coordinate x, without an ε-dependent affine normalization.*

1. *The recurrence coefficients change at rate $(\dot a_n,\dot b_n,\dot c_n)=\pi_n(a_n,\,c_n-a_n,\,-c_n)$.*
2. *If $\int R_xR_yR_z\,d\mu=0$, then*

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\int R_x^\varepsilon R_y^\varepsilon R_z^\varepsilon\,d\mu_\varepsilon \;=\; 2-T_x(y,z)-T_y(x,z)-T_z(x,y),$$
$$T_x(y,z)\;=\;\sum_{k\ge x}g(y,z;k).$$

Proof. Write $R_n^\varepsilon=R_n+\varepsilon Q_n+O(\varepsilon^2)$, with $\deg Q_n\le n$ and $Q_n(x_0)=0$. First-order orthogonality gives $\int Q_n p\,d\mu=p(x_0)$ for $\deg p<n$, hence $Q_n=\sum_{k<n}\pi_k(R_k-R_n)$,

and $\pi_n^\varepsilon = \pi_n\big(1+\varepsilon(2S_n+\pi_n)\big)+O(\varepsilon^2)$ with $S_n = \sum_{k<n}\pi_k$. For (1), differentiate $c_n = \pi_{n-1}\int xR_nR_{n-1}\,d\mu$ and $b_n = \pi_n\int xR_n^2\,d\mu$, using the recurrence, the row sums $a_m+b_m+c_m = x_0$ and $\pi_{n-1}a_{n-1} = \pi_n c_n$; the terms with S_n cancel, and $\dot a_n$ follows from the row sums. For (2), the derivative equals $-R_xR_yR_z(x_0)+\int(Q_xR_yR_z+R_xQ_yR_z+R_xR_yQ_z)\,d\mu$, and $\int Q_xR_yR_z\,d\mu = \sum_{k<x}g(y,z;k) = 1-T_x(y,z)$. ∎

Proposition 9.8 *(Interior points near the symmetric endpoint).* *For every $N\ge 2$ there is $\Lambda_N<\infty$ such that, for $\Lambda>\Lambda_N$ and all sufficiently small $\varepsilon>0$, the measure of $K_{N,1-\Lambda\varepsilon}$ with its top mass lowered by ε has $g(m,n;k)>0$ for all $|m-n|\le k\le \min(m+n,N)$. Consequently, after the affine normalization of Section 2, a relatively open set of arrays near it has property (P).*

Proof. By the model of Lemma 5.2, $K_{N,r}$ has

$$g(m,n;k) = \sum_j H(m,n;j)\binom{j}{s}r^{j-s}(1-r)^s,$$
$$H(m,n;j) = \frac{\binom{m}{j}\binom{N-m}{n-j}}{\binom{N}{n}}, \qquad s = k-m-n+2j,$$

where the sum is over $\max(0,m+n-N)\le j\le\min(m,n)$ with $0\le s\le j$. At $r=1$ only $s=0$ survives, so a triangle coefficient of $K_{N,1}$ vanishes exactly when $m+n+k$ is odd or $m+n+k>2N$. Both deformations are smooth in (r,ε), so only these zeros need first-order positivity.

- If $m+n+k>2N$, all three tails in Lemma 9.7(2) vanish, since $g(y,z;k)=0$ for $k>2N-y-z$. The top-mass rate is therefore $2\pi_k>0$, and the rate of decreasing r is nonnegative because $g\ge 0$ for $r\le 1$.
- If $m+n+k$ is odd and at most $2N+1$, the rate of decreasing r is the $s=1$ term $H(m,n;j)\,j$ with $j=(1+m+n-k)/2$. It is positive, because $j\ge 1$ and $\max(0,m+n-N)\le j\le\min(m,n)$.

These cases cover all zeros. Let $\mathcal{Z}_N$ be the set of triangle triples whose coefficient vanishes at $K_{N,1}$. For $z\in\mathcal{Z}_N$, write A_z for the top-mass rate and B_z for the rate of decreasing r, and define

$$\Lambda_N = \max\Big(\{0\}\cup\Big\{-\frac{A_z}{B_z} : z\in\mathcal{Z}_N,\ A_z<0\Big\}\Big).$$

Every denominator appearing here is positive by the preceding cases. Hence $\Lambda>\Lambda_N$ makes every reference-zero derivative strictly positive, and finiteness of $\mathcal{Z}_N$ gives a common threshold for ε.

Here the measure of $K_{N,r}$ is its probability orthogonality measure. Choose ε so small that $r = 1-\Lambda\varepsilon>0$ and $\varepsilon<(r/(1+r))^N$, so that the lowered top mass remains positive. The affine coordinate $y = (1-\varepsilon)x+\varepsilon = R_1^\varepsilon(x)$ restores the basic recurrence normalization and leaves all product coefficients unchanged. Strict positivity on the triangle gives $b_n = g(1,n;n)>0$ for $1\le n\le N$. Lemma 7.5, with Σ the whole triangle, therefore gives a

relatively open neighborhood with property (P) in the affine space defined by the basic boundary and row-sum equalities. ■

Exact rational computation of the maximum above gives $\Lambda_N = 4, 10, 18, 28, 52, 90, 165$ for $N = 4, \dots, 10$. These finite-size values do not establish an asymptotic growth law or a geometric thickness estimate. At $K_{N,1}$, in the fixed spectral coordinate, Lemma 9.7(1) gives

$$\dot{c}_n = -\frac{n}{N}\binom{N}{n}, \qquad \dot{b}_n = \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}.$$

After the affine normalization restoring $R_1 = x$, the top-mass direction is

$$\delta a_n = \frac{N-n}{N}\Big(\binom{N}{n} - 1\Big), \qquad \delta b_n = 1 + \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}, \qquad \delta c_n = -\frac{n}{N}\Big(1 + \binom{N}{n}\Big).$$

Its binomial dependence on N is not a uniform rule of Definition 9.3, consistently with Theorem 9.4.

Proposition 9.9 *(Families depending on N).* *Let v_N be directions such that a set with property (P) contains $K_{N,1} + \varepsilon v_N$ for all large N and small $\varepsilon > 0$, and let A_N be the alternating sum in Lemma 9.2 for v_N. Then $s_N A_N \geq 0$ for all large N.*

1. *Along either parity class, A_N is not eventually strictly positive and not eventually strictly negative. If A_N has an eventual weak sign along a parity class, it vanishes for all large N of the residue class modulo 4 on which that sign conflicts with the sign of s_N.*

2. *Suppose $\delta b_n = \beta(n/N) + p_n + q_{N-n}$ and $\delta c_n = \gamma(n/N) + p'_n + q'_{N-n}$, with β, γ as in Corollary 9.6 and finitely supported boundary layers p, q, p', q' independent of N. Then $q'_0 = L_{\text{even}} = L_{\text{odd}}$, $p_k = q_k$ and $p'_k + q'_k = -p_k$ for $k \geq 1$.*

Proof. A negative first-order change of $g(N, N; N)$ excludes $K_{N,1} + \varepsilon v_N$ for small ε, so $s_N A_N \geq 0$. Part (1) follows because the sign of s_N alternates along each parity class. In (2), write $D_N f = \sum_{n=0}^{N} (-1)^n \binom{N}{n} f(n/N)$. For large N, $A_N = D_N(\beta + 2\gamma) + P_e(N)$ for even N and $A_N = D_N \beta + P_o(N)$ for odd N, where

$$P_e(N) = q'_0 - L_{\text{even}} + \sum_{k \geq 1} (-1)^k \binom{N}{k} \big[p_k + q_k + 2(p'_k + q'_k)\big],$$

$$P_o(N) = q'_0 - L_{\text{odd}} + \sum_{k \geq 1} (-1)^k \binom{N}{k} (p_k - q_k).$$

These are polynomials in N. A nonzero polynomial has an eventual strict sign, which the remainder $O(\sqrt{N} e^{-\eta N})$ of Lemma 9.5 cannot change, so part (1) forces $P_e \equiv P_o \equiv 0$. The polynomials $1, \binom{N}{1}, \binom{N}{2}, \dots$ are linearly independent, which gives the stated conditions. ■

Thus a separate adjustment of the top coefficient c_N reduces the two bulk equalities to the single equality $\gamma(0) + \beta(1) + \gamma(1) = 0$. For $k \geq 1$, the boundary-layer contributions

satisfy the palindromic first-order relations $\delta b_k = \delta b_{N-k}$ and $\delta a_k = \delta c_{N-k}$; the separate terminal adjustment need not itself be palindromic. Families not covered by part (1) must have A_N change sign, or vanish, infinitely often within a parity class.

Criteria that cover $r < 1$ but not $r = 1$ are thin near $r = 1$.

Proposition 9.10 *(Thinness near the symmetric endpoint).* *Fix $N \geq 2$ and a direction v along which the first-order change $\dot{g}$ of $g(N, N; N)$ at $K_{N,1}$ is negative. There are $r_* < 1$ and $C, c > 0$ such that $K_{N,r} + \varepsilon v$ lacks property (P) whenever $r_* < r < 1$ and $C(1-r)^N \leq \varepsilon \leq c$.*

Proof. The function $G(r, \varepsilon) = g(N, N; N)$ at $K_{N,r} + \varepsilon v$ is a rational function of the recurrence entries whose denominators involve only upward coefficients, which are positive at $K_{N,1}$; so G is smooth near $(1, 0)$. By Lemma 9.1, $G(r, 0) = (1-r)^N$. Put $d = |\dot{g}|$. On a small rectangle around $(1, 0)$, on which the basic sign conditions persist, $\partial_\varepsilon G \leq -d/2$, hence $G(r, \varepsilon) \leq (1-r)^N - d\varepsilon/2 < 0$ for $\varepsilon \geq 4(1-r)^N/d$. ∎

Corollary 9.11 *(Optimal order of the stability radius).* *For $N \geq 2$ let $\rho_N(r)$ be the largest ρ such that every array satisfying (2.1) with $\widetilde{b}_i \geq 0$ and with left side of (5.4) at most ρ has property (P). Then $\rho_N(r)$ is of exact order $(1-r)^N$ as $r \to 1$.*

Proof. Theorem 5.3 gives $\rho_N(r) \geq \eta_N(r) = \frac{1}{2}((1-r)/(32N))^N$ for $r \geq \frac{1}{2}$. For the upper bound, use the terminal direction $\delta c_N = -\operatorname{sgn}(s_N)$, $\delta b_N = \operatorname{sgn}(s_N)$, with all other entries unchanged. By Lemma 9.2 its first-order change is $-|s_N| < 0$, and it moves the full array by exactly ε in the norm of (5.4). Put $t = 1 - r$ and $\varepsilon = Ct^N$ as in Proposition 9.10. For $N \geq 2$ and r near 1, $\varepsilon \leq t/2$ and $\varepsilon \leq r/2$, so $\widetilde{b}_N \geq t/2$ and $\widetilde{c}_N \geq r/2$; every other coefficient is unchanged. The array satisfies the basic conditions and $\widetilde{b}_i \geq 0$, lies at full-array distance $\varepsilon = C(1-r)^N$ from $K_{N,r}$, and lacks property (P). Hence $\rho_N(r) \leq C(1-r)^N$. The restriction $N \geq 2$ is needed: every admissible two-point array with $b_1 \geq 0$ has property (P). ∎

The order $(1-r)^N$ is attained already as $r \to 1$; for fixed r the radius also decays exponentially in N. The deformation that shows this raises the top mass, the reverse of Lemma 9.7, and it is exact rather than first order.

Proposition 9.12 *(Raising the top mass).* *Let $N \geq 1$ and $0 < r < 1$. The measure of $K_{N,r}$ has nodes $x_n = 1 - (1+r)n/N$ and masses $w_n = \binom{N}{n} r^{N-n}/(1+r)^N$. For $\tau > 0$ let μ^τ be this measure with the mass at $x_0 = 1$ raised to $(1+\tau)w_0$. Then μ^τ has*

$$g(N, N; N) = \frac{(1+\tau)^2(I - w_0) + w_0}{(1+\tau)(h - w_0) + w_0}, \qquad I = \big(r(1-r)\big)^N, \quad h = r^N,$$

which is negative exactly when $\tau > \tau_N^* = \big(1 - (1-r^2)^N\big)^{-1/2} - 1$. After the affine normalization of Section 2, its recurrence coefficients differ from those of $K_{N,r}$ by at most $3\tau \max_n w_n$.

Proof. Write π_n and $S_n = \sum_{j<n} \pi_j$ for $K_{N,r}$, and put $\varepsilon = \tau w_0$. The polynomial $(1 + \varepsilon S_n)R_n - \varepsilon \sum_{j<n} \pi_j R_j$ has degree n and equals 1 at x_0. For $\deg p < n$, its integral against

p with respect to $\mu^\tau = \mu + \varepsilon\delta_{x_0}$ is $-\varepsilon p(x_0) + \varepsilon p(x_0) = 0$, because $\sum_{j<n} \pi_j R_j$ reproduces $p(x_0)$. Hence it is the polynomial R_n^τ of μ^τ.

On the support, dual orthogonality gives $\sum_{j\le N} \pi_j R_j(x) = \mathbf{1}_{x=x_0}/w_0$, and in particular $S_N + \pi_N = 1/w_0$. Therefore $R_N^\tau = (1+\tau)R_N - \tau\mathbf{1}_{x_0}$ on the support. By Lemma 9.1 and $\pi_N = r^{-N}$, $\int R_N^3\, d\mu = I$; also $\int R_N^2\, d\mu = h$. Hence

$$\int (R_N^\tau)^3 d\mu^\tau = (1+\tau)^3(I - w_0) + (1+\tau)w_0,$$
$$\int (R_N^\tau)^2 d\mu^\tau = (1+\tau)^2(h - w_0) + (1+\tau)w_0,$$

and their ratio is the stated formula. Since $r > r/(1+r) > r(1-r)$, we have $h > w_0 > I$. So the denominator is positive, and the numerator is negative exactly when $(1+\tau)^2 > w_0/(w_0 - I) = \big(1 - (1-r^2)^N\big)^{-1}$.

For the coefficients, comparing leading coefficients and using detailed balance as in Lemma 9.7 gives

$$a_n^\tau = a_n \frac{1+\varepsilon S_n}{1+\varepsilon S_{n+1}}, \qquad c_n^\tau = c_n \frac{1+\varepsilon S_{n+1}}{1+\varepsilon S_n},$$

and the row sums stay 1. Since $\varepsilon\pi_n = \tau w_n$, each of a_n, b_n, c_n changes by at most τw_n. Put $A_n = 1 + \varepsilon S_n$ and $t_n = \varepsilon\pi_n = \tau w_n$. Since $b_0^\tau = \varepsilon/(1+\varepsilon)$, the affine coordinate is $y = (1+\varepsilon)x - \varepsilon$, and the normalized coefficients are $\widetilde{a}_n = (1+\varepsilon)a_n^\tau$, $\widetilde{b}_n = (1+\varepsilon)b_n^\tau - \varepsilon$, $\widetilde{c}_n = (1+\varepsilon)c_n^\tau$. For $n \ge 1$ we have $A_n \ge 1 + \varepsilon$, and

$$\begin{aligned}
\widetilde{a}_n - a_n &= \varepsilon a_n - (1+\varepsilon)a_n t_n/A_{n+1},\\
\widetilde{c}_n - c_n &= \varepsilon c_n + (1+\varepsilon)c_n t_n/A_n,\\
\widetilde{b}_n - b_n &= -\varepsilon(a_n + c_n) + (1+\varepsilon)t_n\big(a_n/A_{n+1} - c_n/A_n\big).
\end{aligned}$$

Because $a_n, c_n \ge 0$ and $a_n + c_n \le 1$, each absolute change is at most $\varepsilon + t_n \le 2\tau \max_j w_j$. Row zero is restored exactly. This proves, in particular, the stated bound $3\tau \max_j w_j$ for every $\tau > 0$. ∎

Corollary 9.13 *(Exponentially small stability radius).* *Let $0 < r < 1$, and let $\rho_N(r)$ be as in Corollary 9.11. If $(1-r^2)^N \le \frac{1}{2}$ and $3(1-r^2)^N \le (1-r)/N$, then*

$$\rho_N(r) \le 3\tau_N^* \max_n w_n \le 3(1-r^2)^N.$$

In particular $\rho_N(r) = O\big((1-r^2)^N\big)$ as $N \to \infty$, for every fixed $r \in (0,1)$.

Proof. Put $x = (1-r^2)^N$ and $W = \max_n w_n$. For $0 < x \le \frac{1}{2}$,

$$(1+x)^2(1-x) - 1 = x(1 - x - x^2) > 0,$$

so $\tau_N^* < x$. Choose $\tau \in (\tau_N^*, x)$. Proposition 9.12 gives a negative corner coefficient and full-array distance at most $3\tau W < 3x \le (1-r)/N$. Thus $\widetilde{b}_n > 0$ for $n \ge 1$, while $\widetilde{b}_0 = 0$. The exact coefficient formulas and the positive affine scale give $\widetilde{a}_n > 0$ for $n < N$, $\widetilde{c}_n > 0$ for $n > 0$, and all the basic boundary and row-sum equalities. Letting $\tau \downarrow \tau_N^*$ proves $\rho_N(r) \le 3\tau_N^* W \le 3x$. For each fixed $r \in (0,1)$, both smallness conditions hold for all sufficiently large N, since $N(1-r^2)^N \to 0$. ∎

Thus, for every fixed r, no set of arrays with property (P) contains coefficient balls of a radius independent of N around all $K_{N,r}$; Theorem 5.3 gives the lower bound $\rho_N(r) \geq \frac{1}{2}(\min(r, 1-r)/(32N))^N$. The deformation of Proposition 9.12 has the binomial profile w_n and is not a uniform rule. Along the terminal row alone, the first-order change of $g(N, N; N)$ at $K_{N,r}$ has an exact contour-integral representation, obtained as in Lemma 9.2 with $R_N(J)$ having eigenvalues $(-r)^j$. The formal saddle contributions have exponential factor $(1 - r + r^2)/(1 + r)$, which exceeds $1 - r$ exactly when $r > \frac{1}{2}$. For $r > \frac{1}{2}$ this suggests an oscillatory first-order envelope, not a proved threshold for finite perturbations. No asymptotic estimate or uniform control of the nonlinear remainder is proved here. Exact computations (`fixedr_probe.py`) confirm failures of property (P) along the terminal row at sizes down to 10^{-15} for $r = 4/5$, $N = 40$. The comparison at $r = \frac{1}{2}$ concerns only the first-order term. Along the terminal row, $g(N, N; N)$ is a polynomial in the perturbation, since the recurrence for the linearization coefficients divides only by $a_0, \ldots, a_{N-1}$. At $r = \frac{1}{2}$, computations for $N = 30, \ldots, 77$, using maxima over successive six-size windows, give estimated envelope decay factors $0.503, 0.593, 0.640, 0.676$ for the absolute Taylor coefficients of orders $1, 2, 3, 4$, respectively. This is a finite-range computational observation, not an asymptotic statement. Exact terminal-row failures also occur for $r \leq \frac{1}{2}$: at size $6.3 \cdot 10^{-6}$ for $r = \frac{1}{2}$, $N = 120$; at size $3.2 \cdot 10^{-3}$ for $r = \frac{2}{5}$, $N = 100$; and at size $1.6 \cdot 10^{-2}$ for $r = \frac{1}{3}$, $N = 120$.

Computations away from $r = 1$. The explicit fixed-size, one-kink, and log-concave-increment examples below are reproduced by `rigidity_probe.py` and `smallr_probe.py`. Exact rational computations certify the stated finite examples; the separately labelled high-precision calculations provide numerical evidence. No asymptotic conclusion is asserted. Here $g(N, N; N)$ is computed from column N of the product table, an $O(N^2)$-operation recurrence, so sizes in the hundreds are cheap.

- For the rule $\beta = 1$, $\gamma = 0$, exact computation shows that property (P) fails at $r = 4/5$, with perturbation size $1/100$, for every $N = 6, \ldots, 14$, and at $r = 19/20$, with perturbation size $1/1000$, for every $N = 4, \ldots, 14$. The period-four sign pattern at $r = 1$ does not persist in general for $r < 1$: at $r = 4/5$ the first-order derivatives are $512/1875$ for $N = 4$ and $-71936/5859375$ for $N = 8$.

- Below $r = \frac{1}{2}$, the counterexamples we found to shape conditions occur at sizes that grow as r decreases; we did not determine minimal sizes or prove a scaling. For constant-ratio arrays with convex profile, one-kink counterexamples (m unit increments, then $1 + \delta$) have $g(N, N; N) < 0$ in exact arithmetic at

r	$2/5$	$1/3$	$3/10$	$1/4$	$2/7$	$1/5$	$1/10$
N	20	40	40	60	80	100	500

- Convex profiles with log-concave increments (step ratios nonincreasing and at least 1) contain Krawtchouk, every geometric profile and the normalized quadratic profiles $d_\kappa(x) = (x + \kappa x^2)/(1 + \kappa)$, $\kappa \geq 0$, and they exclude the one-kink counterexamples. Yet k steps of ratio $1 + 1/(2N)$ followed by flat increments give $g(N, N; N) < 0$ at

$(r, N, k) = (\frac{1}{2}, 80, 40)$, $(\frac{2}{5}, 32, 22)$ and $(\frac{1}{3}, 90, 62)$ in exact arithmetic. In 80- and 150-digit floating-point arithmetic, which agree but are not exact sign certificates, the same happens at $(\frac{1}{4}, 250, 175)$ and $(\frac{1}{5}, 500, 400)$.

The exact examples above disprove the stated shape conditions at the listed parameters and sizes. The two larger high-precision examples provide additional numerical evidence. These findings establish neither minimal failure sizes nor general conclusions about smooth deformations.

10. Open problems

Inequalities beyond geometric increments. The geometric criterion imposes exact equalities, whereas the perturbation criterion provides small neighborhoods. Theorem 8.1 gives inequalities on the profile that retain the full Krawtchouk range, but it keeps the ratio b_i/c_i constant. Find direct inequalities that also allow a varying ratio, for instance a ratio that increases slowly with i. Exact computations suggest that linearly increasing ratios with log-concave Newton profiles keep property (P), whereas log-concavity of the Newton coefficients of both d and c together with monotone b does not: $d = (0, 25, 55, 91)/91$, $c = (0, 24, 54, 90)/91$ gives $g(3, 3; 3) = -3/2002$. The terminating row and products of degree sum greater than N must remain part of the conclusion.

Sharper stability radii. For fixed N and $0 < r < 1$, determine or estimate the largest coefficient radius for which every admissible stochastic perturbation has property (P). Corollary 9.11 shows that the order $(1-r)^N$ as $r \to 1$ is optimal, and Corollary 9.13 gives $\rho_N(r) = O((1-r^2)^N)$ for each fixed r; the gap to the lower bound $\frac{1}{2}(\min(r, 1-r)/(32N))^N$ of Theorem 5.3 remains open. The bound in Theorem 5.3 comes from a single-term lower margin and a worst-case recurrence amplification. Both estimates discard substantial information. A useful improvement would retain a transparent coefficient test and describe its dependence on both N and the bias parameter.

The symmetric endpoint. At $r = 1$, the reference margin in Lemma 5.2 vanishes because parity forces many coefficients to be zero. Symmetric perturbations preserve the parity zeros, and Theorem 7.6 exploits this for Szwarc's quadratic system when $p_0 < 1/2$. At $p_0 = 1/2$, and for the symmetric Krawtchouk array, further coefficients on the parity support vanish. Determine perturbation directions and structural constraints that preserve positivity there; continuity alone does not control coefficients emerging from those zeros. For every $N \geq 1$, the symmetric Krawtchouk array is a limit of arrays whose product coefficients are strictly positive throughout the triangle support. For $N \geq 2$ this follows from Proposition 9.8; the two-point case is immediate. Theorem 9.4 and Corollary 9.6 impose two boundary equalities on admitted polynomial rules and on the analytic rules satisfying the holomorphic-extension hypothesis specified there. A description of the cone for all N is open.

Monotonicity-type criteria for finite systems. Szwarc's Theorems 3.2 and 3.3 in [6] are open conditions of monotonicity type for infinite systems. For finite systems, property (P) at the boundary point is the hypergroup property of the corresponding birth–death chain, and Miclo [5, Theorem 31 and Proposition 6] adapts the wave-equation argument of Bakry and Huet to give inequality conditions for chains that are symmetric under $n \mapsto N - n$ and sufficiently convex. These conditions do not cover the full Krawtchouk or Szwarc families. For symmetric Krawtchouk they fail when $N > 2$, but the cases $N = 1, 2$ are covered; the Szwarc instance $N = 1$, $p_0 = 1/2$ is the same covered three-point system. By Theorem 7.8, any criterion of this kind for nonsymmetric finite systems, valid including the terminal indices, immediately gives one for symmetric finite systems. Theorem 7.10 supplies such hypotheses for all triples except those whose pairwise sums all exceed N. Theorem 7.12 supplies such hypotheses for all triples under its strengthened conditions (i)–(iii), and Corollary 7.13 contains the arrays (2.4) with $q \geq 2$ but not the Krawtchouk arrays with $N \geq 3$. For constant-ratio arrays, Theorem 8.1 controls the deep triples by inequalities that contain the whole geometric criterion, in particular the Krawtchouk arrays; the remaining task is to allow a nonconstant ratio. Convexity of the profile does not suffice even with a constant ratio and r bounded away from 1: for d with increments 1 at the first seven steps and 6/5 at the remaining seven ($N = 14$), exact arithmetic gives $g(14, 14; 14) < 0$ at $r = 1/2$, and the analogous profile with $N = 20$, twelve unit increments and 21/20 afterwards gives $g(20, 20; 20) < 0$ at $r = 2/5$. Log-convex increments fail as well: fourteen unit increments followed by $(21/20)^t$, $t = 1, \ldots, 4$, give $g(18, 18; 18) < 0$ at $r = 1/2$. These examples show that the stated shape restrictions do not guarantee positivity at the specified biases and sizes. Each failure is witnessed by a negative terminal coefficient $g(N, N; N)$. Section 9 explains part of this: admission at the symmetric endpoint imposes two first-order boundary equalities for the polynomial rules of Theorem 9.4 and the analytic rules of Corollary 9.6, and the computational discussion there gives counterexamples to convex-profile conditions and to convex-profile conditions with log-concave increments at the stated parameters and sizes. Two questions remain. First, Corollary 9.13 gives an exponential upper bound on the admissible coefficient-ball radius around $K_{N,r}$ for every fixed $r \in (0, 1)$, witnessed by raising the top mass. Determine whether analogous upper bounds hold for terminal-row perturbations, and which uniform rules exhibit such thinness. The first-order comparison in the remark after that corollary changes at $r = \frac{1}{2}$. The exact terminal-row failures reported there for $r \leq \frac{1}{2}$ show that higher-order terms matter, so this comparison does not locate a threshold for finite perturbations. Second, decide whether any inequality criterion covers a range $0 < r \leq r_0$ for all N. A criterion given by linear inequalities in the coefficients admits a convex set of arrays, but the set of arrays satisfying property (P) need not be convex, even at fixed (N, r) and with both endpoints admitted by Theorem 2.1. For $q \geq 1$ let $G_{N,q,r}$ be the array (2.4), so that $G_{N,1,r} = K_{N,r}$. For $q > 1$ and $0 < \lambda < 1$ the combination $(1 - \lambda)K_{N,r} + \lambda G_{N,q,r}$ is a stochastic array with constant ratio b_i/c_i and increments proportional to $1 + t\, q^{i-N}$, where $t = \lambda N(q - 1)q^{N-1}/((1 - \lambda)(q^N - 1))$. Exact computation (`convexity_probe.py`) gives $g(N, N; N) < 0$ for this combination at $(r, N, q, t) = (4/5, 10, 10, 1)$, $(3/5, 20, 10, 2)$, $(1/2, 20, 10, 1)$, $(2/5, 60, 10, 2)$, $(1/3, 100, 10, 2)$, $(3/10, 200, 2, 5)$ and $(2/7, 200, 10, 5)$. Com-

putations in 150- and 300-digit arithmetic agree on $g(N, N; N) < 0$ at $(1/4, 500, 3, 10)$; this is not an exact sign certificate. Hence, for each of the seven exactly certified parameter choices above, no sufficient criterion for property (P) whose admitted set is convex, in particular no sufficient criterion defined by a system of linear inequalities in the coefficients, can admit both $K_{N,r}$ and $G_{N,q,r}$. In our searches, the sizes of the examples found tended to increase as r decreased; at $r = 1/5$ no negative corner was found on the tested (N, q, t) grid with $N \le 1000$. Sign conditions on the shape fail in the same way. Write $d_n = b_n + c_n$ and $\rho_n = c_n/d_n$ for $n \ge 1$. For $S_1, S_2 \in \{[0,\infty), (-\infty, 0], \{0\}, \mathbb{R}\}$ and $\rho_{\max} > 0$, consider the stochastic arrays with $b_0 = 0$, a_n decreasing, b_n and c_n nondecreasing, $d_{n+2} - 2d_{n+1} + d_n \in S_1$ for $0 \le n \le N-2$, $\rho_{n+1} - \rho_n \in S_2$ for $1 \le n \le N-1$, and $\rho_n \le \rho_{\max}$. Each such class contains every $K_{N,r}$ with $r \le \rho_{\max}$, and for $S_1 = S_2 = \{0\}$ it consists of these arrays only. For every other choice of (S_1, S_2) and every $\rho_{\max} \ge 83/400$, exact computation (`signpattern_probe.py`) gives an array in the class with $g(N, N; N) < 0$. In the first two examples, normalize the listed increments to sum to one. If S_1 contains a positive number, take constant ratio $1/5$ and eighty unit increments followed by twenty increments $11/10$ ($N = 100$). If S_1 contains a negative number, take constant ratio $1/10$ and seventeen unit increments followed by three increments $1/2$ ($N = 20$). If $S_1 = \{0\}$ and S_2 contains a positive number, take $d_n = n/N$ and $\rho_n = 1/5 + \max(n - 102, 0)/2400$ ($N = 120$). If $S_1 = \{0\}$ and S_2 contains a negative number, take $d_n = n/N$ and $\rho_n = 1/5 - \max(n - 68, 0)/1600$ ($N = 80$). Thus, for every $\rho_{\max} \ge 83/400$, among the classes just defined, only $S_1 = S_2 = \{0\}$ guarantees property (P) for all N. For $\rho_{\max} < 83/400$, these four examples do not settle all sign choices; the cases not excluded by them remain unresolved here. For instance, with $S_1 = \{0\}$ and increasing ratio starting from $\rho = 1/10$, our search found no failure for $N \le 300$. Criteria with conditions on higher differences, with mixed conditions, or with conditions that do not apply at every index are not covered. Szwarc-type certificates were also tested on the deep triples. Floating-point linear-programming searches returned feasible backward-cone certificates for every deep triple in the tested Krawtchouk arrays. For some triples, the joint search for one coefficient-independent weight at $r = 1/5$ and $r = 1/2$ was reported infeasible. These numerical findings concern the specified certificate construction; no exact infeasibility certificates or general exclusion of linear-inequality criteria are supplied. Checkerboard-restricted searches did not certify all deep triples at the tested $r < 1$. The exact marching calculations also produced examples with negative boundary flux. On the tested random perturbed arrays, the solver's certificate-feasibility results agreed with property (P).

Other finite families. Find analogous coefficient regions around other finite orthogonal systems and compare their geometry with the Krawtchouk neighborhoods. An algebra with a positive basis may supply the reference margin, but a recurrence criterion must express the resulting admissible deformations without requiring the unknown product coefficients as input.

AI-use disclosure

OpenAI Codex was used to develop the proof arguments, search and check source passages, construct exact-arithmetic diagnostics, draft and revise this manuscript, and prepare the Lean translation. Separate AI-assisted reviews checked the algebraic proof, boundary cases, normalizations, and quantitative estimates; these reviews are not human peer review. Anthropic Claude was used to analyse Szwarc's Problem 2 against the survey text, to explore discrete hyperbolic certificates numerically, and to develop and draft Sections 7, 8 and 9 together with their exact diagnostics. The kernel recurrence and the operator inequality of Section 8 arose in research consultations with ChatGPT Pro; the closed form of the recurrence for every degree and the exact checks were completed with Claude, which also prepared the Lean formalizations of Theorem 8.1 and of the results of Sections 7 and 9 listed in the section on code and formalization. The review of Section 8 was therefore carried out by the same system that proposed its key steps. These sections were then reviewed in fourteen rounds by ChatGPT Pro (an AI review with independent exact checks, not human peer review); the first round failed, the corrections are recorded in `REVIEW.md`, and for each section the final confirmation round on the corrected text returned PASS. The author set the research objective and retains responsibility for publication decisions and the final mathematical claims. No independent human verification is asserted here. "Machine-verified" refers only to the Lean theorems listed in the section on code and formalization, with the axioms stated there. Other results and the prose remarks are supported by written proofs, AI review and exact numerical checks, but are not machine-verified.

Code and formalization

The companion source tree contains `probe.py` and `verify.py`. The command

```
python3 -u verify.py --max-n 7
```

compares recurrence-matrix multiplication with direct multiplication modulo the terminal polynomial on exact rational inputs, checks the perturbation lower bounds, and replays two rejected coefficient-profile tests. It also checks Theorem 7.1, the margin of Lemma 7.4 and the negative coefficient for $p_0 > 1/2$ on exact Szwarc arrays, the identities of Lemma 7.3 in the graded model, and the lower bound of Theorem 7.6 on rational symmetric perturbations. These computations are diagnostics and do not replace the proofs at arbitrary size. For Section 9, `rigidity_probe.py` checks Lemma 9.2 exactly for $N \leq 20$, the predicted failures of Theorem 9.4, the boundary layers of Proposition 9.9, the region of Lemma 9.5 and the fixed-r computations, and `smallr_probe.py` replays the small-r counterexamples.

The Lean development formalizes the following criteria:

Paper result	Lean declaration
Theorem 2.1: geometric criterion	`Szkraw.geometric_criterion`
Theorem 5.3: perturbation	`Szkraw.perturbation_criterion`
Theorem 7.1: Szwarc's quadratic system	`Szkraw.szwarc_full`, `Szkraw.szwarc_sharp`
Lemma 7.2: edge idempotents	`Szkraw.edge_idempotents`, `Szkraw.cube_model`
Lemma 7.3: three-term action	`Szkraw.three_term_action`, `Szkraw.Uvec_top`
Lemma 7.4: margin for the quadratic system	`Szkraw.szwarc_margin`
Lemma 7.5: stability of a reference with a margin	`Szkraw.stability`
Theorem 7.6: symmetric criterion	`Szkraw.symmetric_criterion`
Corollary 7.7: symmetric systems outside both families	`Szkraw.symmetric_new`
Theorem 7.8: quadratic decomposition	`Szkraw.decomp_positivity`, `Szkraw.decomp_converse`, `Szkraw.even_block`, `Szkraw.odd_block`, `Szkraw.evenComp_admissible`, `Szkraw.oddComp_admissible`
Corollary 7.9: parity-split criterion	`Szkraw.parity_split`, `Szkraw.parity_split_sharp`, `Szkraw.ps_even_params`, `Szkraw.ps_odd_params`
Theorem 7.10: low products	`Szkraw.low_products`
Corollary 7.11: convex profiles	`Szkraw.convex_low`, `Szkraw.geometric_convex`
Theorem 7.12: all products	`Szkraw.all_products`
Corollary 7.13: heavy top step	`Szkraw.heavy_top`
Lemma 7.14: reversal identity	`Szkraw.reversal_identity`
Theorem 7.15: deep reduction	`Szkraw.deep_formula`, `Szkraw.top_expansion`, `Szkraw.deep_reduction`
Theorem 7.16: palindromic criterion	`Szkraw.palindromic`
Corollary 7.17: monotone palindromic systems	`Szkraw.palin_mono`, `Szkraw.palin_mono_kraw`
Theorem 8.1: log-concave Newton criterion	`Szkraw.logconcave_criterion`
Remark after Theorem 8.1: two-parameter and Bernstein coefficients	`Szkraw.lc_twoParam`, `Szkraw.lc_bernstein`, `Szkraw.constantRatio_twoParam`
Lemma 9.1: corner value	`Szkraw.corner_value`
Lemma 9.2: corner identity	`Szkraw.corner_identity`
Theorem 9.4: forced equalities for uniform criteria	`Szkraw.forced_equalities`, `Szkraw.forced_vanishing`
Lemma 9.5: Nörlund–Rice bound	`Szkraw.norlund_rice`
Corollary 9.6: analytic rules	`Szkraw.analytic_forced_equalities`, `Szkraw.analytic_rule_small`, `Szkraw.expRule_analytic`, `Szkraw.expRule_L`, `Szkraw.expRule_deriv`
Proposition 9.9: families depending on N	`Szkraw.families_sign`, `Szkraw.families_weak_sign`, `Szkraw.families_no_strict_sign`, `Szkraw.families_boundary_layers`
Lemma 9.7: lowering the top mass	`Szkraw.lower_measure`, `Szkraw.lower_rates`, `Szkraw.lower_triple`
Proposition 9.8: interior points	`Szkraw.interior_measure`, `Szkraw.interior_open`

The first includes a positive finite orthogonality measure, distinct support points, the terminal relation, and all finite product identities. The second includes the explicit radius, positive triangle margin, and exact zeros outside the triangle. The entries for Section 7 state nonnegativity of the coefficients $g(m,n;k)$: for Theorem 7.10 and Corollary 7.11 on all triples with two indices of sum at most N, and for Theorems 7.12, 7.15 and 7.16 and Corollaries 7.13 and 7.17 on all triples. Lemma 7.14 and the identities of Theorem 7.15 are stated for the matrices $R_m(J)$ and $R'_t(J) = \Pi R'_t(\Pi J \Pi)\Pi$. Theorem 7.1 is proved in Lean through Corollary 7.9 with $q = 1$, and Lemma 7.4 through the decomposition, whose components are Krawtchouk arrays; the symmetry of its measure holds for every finite realization of an array satisfying (2.1) with $b \equiv 0$ (`Szkraw.realization_symmetric`). In Corollary 7.7, "not of the form (2.4)" is formalized as the failure of the hypotheses of Theorem 2.1 and as disagreement with every array (2.4) in the Lean parametrization; its last sentence follows from the explicit coefficient box of positive radius in `Szkraw.symmetric_criterion`; openness and dimension are not stated separately in Lean. For Corollary 7.9, when $N \geq 2$, Lean proves that both normalized components satisfy the hypotheses of Theorem 2.1 and are the arrays (2.4) with parameters (N, q, r) and $(N-1, q, r/q)$ (`Szkraw.ps_even_params`, `Szkraw.ps_odd_params`); `Szkraw.parity_split` also covers $N = 1$, where the odd component is the constant polynomial alone. In Theorem 7.8 the identity for $g(2a, 2b+1; 2c+1)$ is stated with the unnormalized odd component, which has the same product coefficients (`Szkraw.product_normalize`). The finite realization holds for every array satisfying (2.1) (`Szkraw.finite_realization`). The proofs formalize the cone argument and the identity $g(m,n;k)\int R_k^2\,d\mu = g(m,k;n)\int R_n^2\,d\mu$ in matrix form. The entry for Theorem 8.1 includes the same finite realization as the first and the nonnegativity of every coefficient $g(m,n;k)$; its proof formalizes Lemmas 8.3 and 8.4, the operator induction, and Lemma 8.2 in converse form: the solution of (8.1) yields the linearization coefficients. The remark after the proof of Theorem 8.1 is formalized as identities for all real u and v (two-parameter form, `Szkraw.lc_twoParam`) and for all real r (Bernstein form, `Szkraw.lc_bernstein`, whose coefficients B_s are $\binom{D}{s}$ times the Bernstein coefficients); `Szkraw.constantRatio_twoParam` identifies the arrays of Theorem 8.1 with this family, and the proof formalizes the operator induction coefficientwise in t. All of them have compiled proofs without added axioms or admitted goals; the axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. Lemma 7.2 is stated for every commutative real algebra with elements $y_1, \dots, y_N$ satisfying $y_j^2 = (1-r)y_j + r$ whose squarefree monomials form a basis, which determines A up to isomorphism, and for $r \neq -1$; `Szkraw.cube_model` shows that the real functions on $\{1, -r\}^N$ form such an algebra. Its basis assertion is stated as linear independence of the $f_l y_S$ together with equality of their span and the ideal $f_l A$. Lemma 7.3 is stated for $N \geq 1$, $p_0 < 1$ and $r = p_0/(1-p_0)$, in every commutative real algebra satisfying these relations, without a basis hypothesis; its recurrence is an equality in $A \times A^N$, which restricts to V, and `Szkraw.Uvec_top` states $U_{2N+1} = 0$ separately. Lemma 9.2 is stated for $N \geq 2$ as the derivative at $\varepsilon = 0$ of $g(N,N;N)$ along $K_{N,1} + \varepsilon v$. Its Lean proof replaces the beta integrals by an explicit matrix: the derivative X of $R_N^{\varepsilon}(J_\varepsilon)$ satisfies $[J,X] = [\Pi, E]$ and $e_0^{\mathsf{T}} X = 0$, and $Z_{mn} = \binom{N}{n}(-1)^{n+s}/(2\binom{N-1}{s})$ for $m+n = 2s+1$, $Z_{mn} = 0$ for $m+n$ even, satisfies $ZJ - JZ = e_N e_N^{\mathsf{T}}$ on the columns $n \geq 1$; hence the derivative equals $\operatorname{tr}((Z\Pi - \Pi Z)E)$,

which is evaluated entry by entry. In Theorem 9.4 the set of arrays is a family of sets indexed by N whose members satisfy $g(m,n;k) \geq 0$, admission is Definition 9.3, and the second statement is the vanishing of the derivative for every $N \geq 1$ above the degrees. Lemma 9.5 is stated with $U(x) = \int_0^1 \log|x-t|\,dt$ (`Szkraw.Ulog`) for f complex differentiable on an open set containing Ω, as a bound $C\sqrt{N}\,e^{-\eta N}$ for every $N \geq 1$; the cycle of its proof is the boundary of a finite union of squares of an axis-parallel grid with mesh $1/M$ for a positive integer M and origin $\sqrt{2}(1+i)$, so that no node k/N lies on a grid line (`Szkraw.residue_identity`), and the Riemann-sum and Stirling estimates are `Szkraw.riemann_lower` and `Szkraw.factorial_le_stirling`. In Corollary 9.6 and Proposition 9.9 (2), a rule on $[0,1]$ is analytic on an open set containing Ω if it is the restriction of a function complex differentiable there (`Szkraw.AnalyticOnOmega`); the second statement of Corollary 9.6 is the bound $|s_N|\,C\sqrt{N}\,e^{-\eta N}$ on the derivative for every $N \geq 2$, and the example is stated for every $N \geq 2$. In Proposition 9.9 (2) the boundary layers vanish beyond a fixed index K, the formulas for δb_n and δc_n are assumed for $1 \leq n \leq N-1$ and $1 \leq n \leq N$ and all large N, and the hypothesis is that of the first statement; The first statement of Corollary 9.6 is obtained by specializing Proposition 9.9 (2) to zero boundary layers. In Proposition 9.10 the direction is given in the coordinates $b_1,\dots,b_{N-1},c_1,\dots,c_N$ of Section 9, the hypothesis is that the derivative of $g(N,N;N)$ at $\varepsilon = 0$ is negative, and the conclusion is that $K_{N,r} + \varepsilon v$ satisfies (2.1) and has $g(N,N;N) < 0$; `Szkraw.productMatrix_contDiffAt` gives the smoothness used in its proof. In Corollary 9.11, $\rho_N(r)$ is the supremum of the radii with the stated property, and exact order means $c_1(1-r)^N \leq \rho_N(r) \leq c_2(1-r)^N$ for all r in an interval $(r_0,1)$. Lemma 9.7 is stated for the spectral measure of every array satisfying (2.1), whose node $x_0 = 1$ exists by `Szkraw.spectral_top_node`; the polynomials of μ_ε are identified by orthogonality and their value 1 at x_0 (`Szkraw.lower_measure`), and both parts are derivatives at $\varepsilon = 0$. In Proposition 9.8, Λ_N is an explicit finite sum bounding the absolute ratios of the two first-order rates with nonzero r-rate (`Szkraw.lamN`), the lowered measure is identified by orthogonality and normalization, and the neighborhood is a coefficient box of some positive radius around the normalized array; the proof uses an explicit formula for the coefficients of $K_{N,r}$ (`Szkraw.kraw_g_formula`). In Proposition 9.12, Lean states that the normalized array satisfies (2.1), that its polynomials are orthogonal for the raised measure transported to $y = (1+\varepsilon)x - \varepsilon$ and equal 1 at the top node, and then the corner formula, its sign and the coefficient bound; the proof uses the polynomials $(1+\varepsilon S_n)R_n - \varepsilon\sum_{j<n}\pi_j R_j$ for any finite system (`TopMass.lean`). Every numbered result of Section 9 is formalized, and so are three consequences stated in its text: no set with property (P) admits all rules of an open set of polynomial rules of degree at most d, with rules given by coefficient vectors (`Szkraw.no_open_rule_set`); the single equality $\gamma(0)+\beta(1)+\gamma(1) = 0$ and the palindromic relations for the boundary layers after Proposition 9.9 (`Szkraw.boundary_layers_palindromic`); and the absence of coefficient balls of a radius independent of N around all $K_{N,r}$, in the sense of $\rho_N(r)$, after Corollary 9.13 (`Szkraw.radius_no_uniform`). The other prose remarks of Section 9, including the description of Ω and the computations away from $r = 1$, are not formalized.

The companion code is available at https://github.com/skypher/szkraw; the fixed

revision cited in [8] identifies the reviewed proof and audit code. Run `make lean` to build the development and `make axioms` to build and audit the principal statements. The latter invokes `lean/CheckAxioms.lean`, which rejects dependencies outside `propext`, `Classical.choice`, and `Quot.sound` and tests rejection of unexpected axioms. The toolchain and dependencies are pinned in `lean/lean-toolchain` and `lean/lake-manifest.json`. The declaration-by-declaration correspondence, auxiliary formalization coverage, and verification commands are recorded in `RESEARCH.md` at that revision. Historical attribution and the open problems are prose claims, not machine-verified statements.

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